1. IIT-JEE Syllabus

Rates of Chemical reactions, Order of reactions, Rate constant, Effect of concentration and Temperature, Kinetics of first–order reactions, Arrhenius Equation, Kinetics of radioactive disintegrations. Stability of nuclei with respect to neutron proton ratio, carbon dating, brief discussion of fission and fusion reactions.

2. Introduction

Chemical Kinetics is the branch of science that deals with rate of reaction, factors affecting the rate of reaction and reaction – mechanism.

Different reactions occur at different rate. In fact a chemical reaction involves redistribution of bonds — breaking of bond(s) in the reactant molecule(s) and making of bonds in the product molecule(s). The rate of a chemical reaction actually depends upon the strength of the bond(s) and number of bonds to be broken during the reaction. It takes longer time for the reactant molecules to acquire higher amount of energy which they do by collision. Hence reactions involving strong bond – breaking occur at relatively slower rate while those involving weak bond – breaking occur at relatively faster rate. On the basis of rate, reactions are classified as.

i) Instantaneous or extremely fast reactions i.e. reactions with half-life of the order of fraction of second.

ii) Extremely slow reactions i.e. reactions with half-life of the order of years.

iii) Reactions of moderate or measurable rate.

Ionic reactions are instantaneous. If a drop of silver nitrate solution is added to a solution of the chloride salt of any metal or solution of HCl, a white precipitate of silver chloride appears within twinkling of eye. This is because of the fact that in aqueous solution an ionic compound exists as its constituent ions. No bond needs to be broken during the reaction. Hence reaction takes no time to complete. The half life period of an ionic reaction is of the order of 10^{-10} s.

$$Na^+ + Cl^- + Ag^+ + NO_3^- \longrightarrow AgCl\downarrow + Na^+ + NO_3^-$$

Free radicals being very unstable (reactive) due to the presence of unpaired electron, reactions involving free radicals also occur instantaneously. Thus, the reactions, are instantaneous.

$$^\bullet CH_3 + Cl_2 \longrightarrow CH_3Cl + {}^\bullet Cl$$
$$^\bullet CH_3 + {}^\bullet CH_3 \longrightarrow H_3C - CH_3$$

Some molecular reactions involving reactant(s) containing odd electron completes within a fraction of second. The fastness of such reactions is attributable to the tendency of the odd electron molecule (paramagnetic in nature) to transform into stable spin-paired molecule (diamagnetic) by dimerisation. An example of such reaction is the dimerisation of nitrogen dioxide into nitrogen tetraoxide as mentioned below.

$$^\bullet NO_2 + {}^\bullet NO_2 \longrightarrow N_2O_4$$

There are some molecular reactions which are known to be extremely slow. Their half-lives are of the order of several years. Some examples of the type of reactions are as given below:

$$4Fe + xH_2O + 3O_2 \longrightarrow 2Fe_2O_3.xH_2O$$
$$[Cr(H_2O)_6]^{3+} + I^- \longrightarrow [Cr(H_2O)_5I]^{2+} + H_2O$$

Note that the first reaction given above is called "rusting of iron". The second one is not ionic reaction as it appears at the first sight. Here in this reaction it is the co-ordinate bond between central metal ion i.e. Cr^{3+} (acceptor) and water molecule (donor) that broken and covalent bond between Cr^{3+} and I^- that is formed. The half-life of this reaction is in years.

Most molecular reactions especially organic reactions occur at measurable rate. The half-life of such reactions are of the order of minutes, hours, days. Examples of such reactions are numerous. Some of these are given below.

$$CH_3COOC_2H_5 + H_2O \xrightarrow{H^+} CH_3COOH + C_2H_5OH$$

$$\underset{\text{(Sucrose)}}{C_{12}H_{22}O_{11}} + H_2O \xrightarrow{H^+} \underset{\text{(Glucose)}}{C_6H_{12}O_6} + \underset{\text{(Fructose)}}{C_6H_{12}O_6}$$

$$H_2O_2\ (aq) \longrightarrow H_2O + \frac{1}{2}O_2 \uparrow$$

$$2N_2O_5 \longrightarrow 4NO_2 + O_2 \uparrow$$

$$NH_4NO_2\ (aq) \longrightarrow 2H_2O + N_2 \uparrow$$

In Chemical Kinetics we deal with the rates of only those reactions which occur with measurable rate i.e. which are neither too fast nor too slow. These days rates of fast reactions are also determined using lasers.

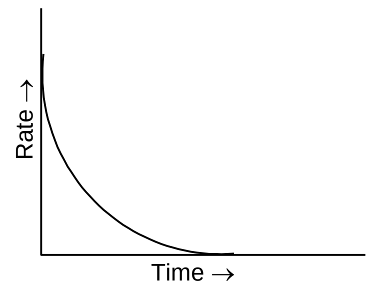
2.1 Rate of Reaction

With the progress of reaction the concentration of reactants decreases while that (those) of product(s) increases. If the reaction occurs fastly, the concentration (mole L^{-1}) of reactant will decrease rapidly while that of product will increase rapidly. Thus, the rate of reaction is defined as the rate at which the concentration of reactant decreases or alternatively, the rate at which concentration of product increases. That is, the change in concentration of any of the substance (reactant / product) per unit time during the reaction is called rate of reaction. If Δ C be the change in concentration of any reactant or any product during the time interval Δt then the rate may given as

$$\text{Rate} = \pm \frac{\Delta C}{\Delta t}$$

(–ve sign applies in the case of reactant whose concentration goes on decreasing with time and +ve sign applies in the case of product whose concentration goes on increasing with time).

However, the rate of reaction is not uniform. With the passage of time the concentration(s) of reactant(s) goes on decreasing and hence according to Law of Mass Action, the rate of reaction goes on decreasing. Infact, the rate of reaction decreases moment to moment. This is shown in the graph of rate vs time curve of a reaction.



Rate varies from moment to moment so rate of reaction has to be specified at a given instant of time called instantaneous rate or rate at any time t. This is as defined below.

$$r_{inst} \text{ or } r_t = \pm\frac{dc}{dt}$$

Where dc is the infinitesimal change in concentration during infinitesimal time interval dt after time t i.e. between t and t + dt. The time interval dt being infinitesimal small, the rate of reaction may be assumed to be constant during the interval.

The rate of reaction expressed as $\pm\Delta C/\Delta t$ is actually the average rate with time interval considered.

For the reaction: $2N_2O_5 \longrightarrow 4NO_2 + O_2$

The rate of reaction at any time t may be expressed by one of the following.

$$\frac{-d[N_2O_5]}{dt}, +\frac{d[NO_2]}{dt} \text{ or } +d\frac{[O_2]}{dt}$$

Where square bracket terms denote molar concentration of the species enclosed.

The above three rates are not equal to one another as is evident from the stoichiometry of the reaction. For every mole of N_2O_5 decomposed, 2 moles NO_2 and 1/2 mole of O_2 will be formed. Obviously, the rate of formation of NO_2 will be four times that of O_2 and it is twice the rate of consumption of N_2O_5. Thus, their three rates are interrelated as,

$$+\frac{d[NO_2]}{dt} = 2\left\{-\frac{d[N_2O_5]}{dt}\right\} = 4\left\{+\frac{d[O_2]}{dt}\right\}$$

Dividing through out by 4 ,

$$\frac{1}{2}\left\{-\frac{d[N_2O_5]}{dt}\right\} = \frac{1}{4}\left\{+\frac{d[NO_2]}{dt}\right\} = +\frac{d[O_2]}{dt}$$

Thus, rate of reaction expressed in terms of the various species involved in a reaction will be equal to one another if each of there is divided by the stoichiometric constituent of the species concerned. Rate divided by stoichiometric coefficient may be called as rate per mole.

The Kinetic experiment has shown that the rate of reaction mentioned above increases same number of times as the number of times the concentration of N_2O_5 is increased. That is, rate is doubled by doubling the concentration of N_2O_5. This may be mathematically expressed as

$$\text{Rate} \propto [N_2O_5]$$
$$\text{or Rate} = k\,[N_2O_5]$$

Where k is the proportionality constant called rate constant, velocity coefficient or specific reaction rate of the reaction, and is a constant for a given reaction at a given temperature. Equation 1 showing the concentration – dependence of rate is called rate law of the reaction. Expressing rate in terms of the rate of change of concentration of various species, equation may be put as

$$-\frac{d[N_2O_5]}{dt} = k'\,[N_2O_5]$$
$$+\frac{d[NO_5]}{dt} = k''\,[NO_2]$$
$$+\frac{d[O_2]}{dt} = k'''\,[O_2]$$

when k', k'' and k''' are the rate constants of the reaction. These three rate constants are inter-related.

$$\frac{k'}{2} = \frac{k''}{4} = k'''$$

Thus, for a reaction represented by the general equation

$$aA + bB \longrightarrow cC + dD$$

$$\frac{1}{a}\left\{-\frac{dC_A}{dt}\right\} = \frac{1}{b}\left\{-\frac{dC_B}{dt}\right\} = \frac{1}{c}\left\{+\frac{dC_C}{dt}\right\} = \frac{1}{d}\left\{+\frac{dC_D}{dt}\right\}$$

We have,

$$\frac{k^{I}}{a} = \frac{k^{II}}{b} = \frac{k^{III}}{c} = \frac{k^{IV}}{d}$$

where

$$k^{I}, k^{II}, k^{III} \text{ and } k^{IV}$$

are the rate constants of the reaction when its rate is expressed in terms A, B, C and D, respectively.

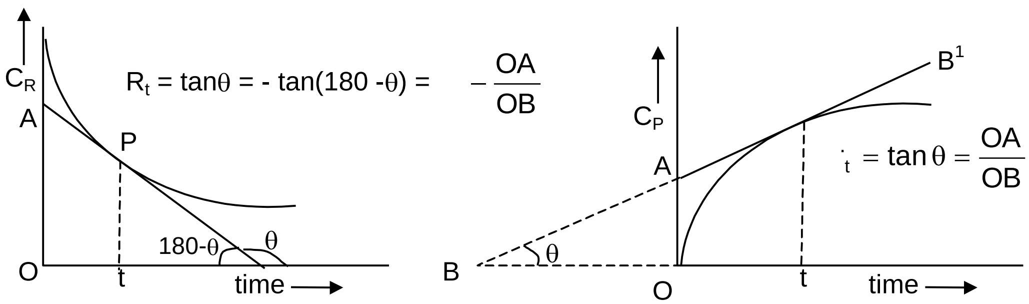
The rate of reaction at any time t is determined in the following way,

i) Concentration of any of the reactants or products which ever may be convenient is determined at various time intervals.

ii) Then concentration vs time curve is drawn.

iii) A tangent is drawn at the point p of the curve which corresponds to the time t at which rate is to be determined.

iv) The slope of the tangent gives the rate of reaction at the required time as shown below.



(C_R and C_P denote concentration of reactant and product respectively)

Unit of rate = $\frac{\text{Unit of concentration}}{\text{Unit of time}}$ = Concentration time^{-1} i.e. mole L^{-1} S^{-1}

2.2 Molecularity

A chemical reaction that takes place in one and only one step i.e., all that occurs in a single step is called elementary reaction while a chemical reaction occurring in the sequence of two or more steps is called complicated reaction. The sequence of steps through which a complicated reaction takes place is called reaction – mechanism. Each step in a mechanism is an elementary step reaction.

The molecularity of an elementary reaction is defined as the minimum number of molecules, atoms or ions of the reactants(s) required for the reaction to occur and is equal to the sum of the stoichiometric coefficients of the reactants in the chemical equation of the reaction. Thus, the molecularity of some elementary reactions are as mentioned below.

Elementary reactions	**Molecularity**
$PCl_5 \rightleftharpoons PCl_3 + Cl_2$	1
$H_2 + I_2 \rightleftharpoons 2HI$	1 + 1 = 2

Reactions with molecularity equal to one, two, three etc., are called unimolecular, bimolecular, termolecular, etc., respectively.

A complicated reaction has no molecularity of its own but molecularity of each of the steps (elementary reactions) involved in its mechanism.

For example, consider the reaction: $2NO + 2H_2 \rightarrow N_2 + 2H_2O$, which is complicated reaction and takes place in the sequence of following three steps.

i) $NO + NO \rightleftharpoons N_2O_2$ (fast and reversible)

ii) $N_2O_2 + H_2 \xrightarrow{\text{r.d.s.}} N_2O + H_2O$ (slow)

iii) $N_2O + H_2 \longrightarrow N_2 + H_2O$ (fast)

The molecularity of each step in the mechanism is two, so what we say is that the reaction takes in the sequence of three steps each of which is bimolecular. There is another view also. According to which molecularity of a complicated reaction is taken to be equal to the molecularity of the slowest step i.e. rate – determining step (r.d.s.) in the mechanism.

For example, the reaction

$$ROH + HCl \xrightarrow{ZnCl_2} RCl + H_2O$$

is said to be unimolecular nucleophilic substitution (S_N1) since the reaction occurs in the sequence of following three steps and the slowest step i.e. r.d.s. is unimolecular.

i) $R\ddot{\underset{\cdot\cdot}{O}}H + HCl \rightleftharpoons R\overset{\oplus}{\underset{\cdot\cdot}{O}}H_2 + :Cl^-$ (fast)

ii) $R\overset{\oplus}{\underset{\cdot\cdot}{O}}H_2 \rightleftharpoons R^{\oplus} + H_2O$ (slow)

iii) $\overset{\oplus}{R} + :Cl^- \longrightarrow RCl$ (fast)

Reactions of higher molecularity (molecularity > 3) are rare. This is because a reaction takes place by collision between reactant molecules and as number of reactant molecules i.e. molecularity increases the chance of their coming together and colliding simultaneously decreases.

2.3 Order of Reaction

The mathematical expression showing the dependence of rate on the concentration(s) of reactant(s) is known as rate-law or rate-expression of the reaction and sum of the indices (powers) of the concentration terms appearing in the rate law as observed experimentally is called order of reaction. To understand what is order of reaction, consider the reaction:

$$2NO(g) + 2H_2(g) \longrightarrow N_2(g) + 2H_2O\ (g)$$

Kinetic experiment carried out at 1100 K upon this reaction has shown following rate data.

Expt. No.	[NO] (mole dm^{-3})	[H_2] (mole dm^{-3})	Rate (mole dm^{-3}s^{-1})
1.	5×10^{-3}	2.5×10^{-3}	3×10^{-5}
2.	1.0×10^{-2}	2.5×10^{-3}	1.2×10^{-4}
3.	1.0×10^{-2}	5.0×10^{-3}	2.4×10^{-4}

From the Expt. No.1 and 2, it is evident that rate increases 4 fold when concentration of NO is doubled keeping the concentration of H_2 constant i.e.

Rate $\propto [NO]^2$ when [H_2] is constant again from Expt. No.2 and 3, it is evident that when concentration of H_2 is doubled keeping the concentration of NO constant, the rate is just doubled i.e.

$$\text{Rate} \propto [H_2] \text{ when [NO] is constant}$$

From Expt. (1) and Expt. (3), the rate increases 8-fold when concentrations of both NO and H_2 are doubled simultaneously i.e.

$$\text{Rate} \propto [NO]^2\,[H_2]$$

This is the rate-law of reaction as observed experimentally. In the rate law, the power of nitric oxide concentration is 2 while that of hydrogen concentration is 1. So, order of reaction w.r.t. NO is 2 and that w.r.t. H_2 is 1 and overall order is 2 + 1 i.e. 3.

Note that the experimental rate law is not consistent with the stoichiometric coefficient of H_2 in the chemical equation for the reaction. This fact immediately suggests that the reaction is complicated and it does not occur in single step as written. In order to explain the rate law, following mechanism has been proposed.

i) $NO + NO \rightleftharpoons N_2O_2$ (fast and reversible)

ii) $N_2O_2 + H_2 \longrightarrow N_2O + H_2O$ (slow)

iii) $N_2O + H_2 \longrightarrow N_2 + H_2O$ (fast)

Let us see how this mechanism corresponds to the rate law as found experimentally and mentioned above.

The step II being the slowest step is the rate-determining step. Thus, rate of overall reaction or rate of formation of N_2, will be equal to the rate of step II or rate of formation of H_2O. So, we have according to Law of Mass Action.

Rate of overall reaction = Rate of step II = $k\,[N_2O_2]\,[H_2]$

Where k = rate constant of step II.

N_2O_2 being intermediate for the overall reaction, its concentration has to be evaluated in terms of the concentration(s) of reactant(s) and this can be done by applying Law of Mass Action upon the equilibrium of Step I. Thus,

$$K_C = \frac{[N_2O_2]}{[NO]^2}$$

$$\text{or } [N_2O_2] = K_C[NO]^2$$

where K_C = equilibrium constant of Step II. Putting this value of concentration of N_2O_2 in the above rate expression, we get

Rate reaction = $k \cdot K_c \cdot [NO]^2\,[H_2]$

or Rate of reaction = $k^1[NO]^2\,[H_2]$

Rate of reaction $\propto [NO]^2\,[H_2]$

Where $k \cdot K_c = k^1$ = another constant, rate constant of overall reaction.

Note that from the knowledge of any two out of k, K_c and k^1, the rest one may be calculated.

We are again turning to our topic "order". In general, if rate law of a reaction represented by the equation.

$$aA + bB \longrightarrow \text{Products}$$

is experimentally found to be as follows:

$$\text{Rate} \propto [A]^m\,[B]^n$$

Then order w.r.t. A = m

order w.r.t. B = n

overall order = m + n

It is to be noted that 'm' may or may not be equal to 'a' and similarly. 'n' may or may not be equal to 'b'. m and n are experimental quantities and their values which really depend on the reaction-mechanism and experimental condition, may not be predicted by just seeing the chemical equation of the reaction. Reactions with some kind of chemical equations may differ in this rate laws and hence order. An example of this is as follows.

Reactions	**Rate Law**	**Order**
$2N_2O_5 \longrightarrow 4NO_2 + O_2$	Rate $\propto [N_2O_5]$	1
$2NO_2 \longrightarrow N_2O_2 + O_2$	Rate $\propto [NO_2]^2$	2

Order of reaction may also be defined as follows.

Number of molecules of the reactant(s) whose concentration changes during the chemical change is called order of reaction.

For example, the reaction

$$CH_3COOC_2H_5 + H_2O \xrightarrow{H^+} CH_3COOH + C_2H_5OH$$

is a bimolecular reaction but its order is 'one'. This is because during the reaction only the concentration of ester decreases with time. The concentration of water in the reaction mixture (usually a dilute aqueous solution of ester mixed with dilute aqueous acid) being in large excess as compared to ester, does not decrease appreciably or measurably during the reaction.

The first order behaviour of this reaction can be derived in the following way.

Applying Law of Mass Action upon the above reaction.

Rate $\propto [CH_3COOC_2H_5]\,[H_2O]$

or Rate $= k[CH_3COOC_2H_5]\,[H_2O]$

Where k is the rate constant of the above bimolecular reaction.

Since concentration of water remains practically constant. So,

$K[H_2O] = k^1$ = another constant or observed rate constant of the reaction.

So, Rate = $k^1 [CH_3COOC_2H_5]$

This is first-order kinetics i.e. order in respect of ester is '1' and that in respect of water is 'zero'. The reaction is an example of pseudounimolecular (or pseudo first order). Thus, a second order reaction conforms to the first order if out of the reactants one is present in large excess and the reaction is called pseudounimolecular.

Suppose we have a reaction:

$2A + B \longrightarrow$ Products

With rate law

Rate $\propto [A]^2 [B]$ ($\therefore$ order = 2 + 1 = 3)

It B is taken in large excess as compared to A, their reaction will obey the kinetics.

Rate $\propto [A]^2$ ($\because$ [B] is constant)

So,

Order w.r.t. A = 2

Order w.r.t. B = 0

Overall order = 2 + 0 = 2

If A is taken in large excess as compound to B then reaction will obey the kinetics.

Rate $\propto [B]$ ($\because$ [A] is constant)

So,

Order w.r.t. A = 0

Order w.r.t. B = 1

Overall order = 0 + 1 = 1

If both A and B are taken in large excess, can you say what will be the order? Some of you may tell that order will be zero. This is absolutely wrong. When both A and B are in large excess, then there will be appreciable damage in the concentrations of both of them and hence order will be '3'.

Reactions are classified on the basis of order as an, first, second, third order etc. according as their order equal to 0, 1, 2and 3 etc. respectively.

Difference between Order and Molecularity

i) Order is an experimental property while molecularity is the theoretical property.

ii) Order concerns with kinetics (rate-law) while molecularity concerns with mechanism.

iii) Order may be any number, fractional, integral or even zero whereas molecularity is always an integer excepting zero.

Example

Reactions	Rate Law	Order
$CH_3CHO \longrightarrow CH_4 + CO$	Rate $\propto [CH_3CHO]^{3/2}$	1.5
$NH_3 \longrightarrow \frac{1}{2}N_2 + \frac{3}{2}H_2$	Rate $\propto [NH_3]^0$	0
$2HI \longrightarrow H_2 + I_2$	Rate $\propto [HI]^0$ i.e. Rate = k	0

Note that a zero order reaction is one in which concentration of reactant remains throughout constant and as such rate of reaction remains throughout constant equal to the rate constant.

iv) Order may change with change in experimental condition while molecularity can't.

Example

This reaction follows 1st order kinetics at high gas pressure and 2nd order kinetics at low gas pressure of cyclopropane.

3. Kinetics of First Order Reaction

A first order reaction is one whose rate varies as 1st power of the concentration of the reactant i.e. the rate increases as number of times as the number of times the concentration of reactant is increased.

Let us consider a unimolecular first order reaction represented by the general equation.

A ——— Product		
a	x = 0	… t = 0
a - x	x = x	… t = t

The initial concentration of A is a mole L^{-1} and its concentration after any time t is (a – x) mole L^{-1}. This means during the time interval t, x mole L^{-1} of A has reacted.

The rate of reaction at any time t is given by the following first – order kinetics.

$$-\frac{d(a-x)}{dt} \propto (a-x) \text{ or } \frac{d(x)}{dt} \propto (a-x)$$

or $\frac{dx}{dt} = k\,(a-x)$

($\frac{da}{dt} = 0 \because$ a has a given value for a given expt.)

where k is the rate constant of the reaction.

$$\frac{dx}{a-x} = kdt$$

This is differential rate equation and can be solved by integration.

$$\int \frac{dx}{a-x} = k\int dt$$

or $-\ln(a-x) = k.t + C$ …(1)

Where C is integration constant.

The constant C can be evaluated by applying the initial condition of the reaction i.e. when t = 0, x = 0. Putting these in equation 1, we get

$C = -\ln a$

Putting the value of C in equation 1, we get

$$-\ln(a-x) = k.t. - \ln a \text{ or } k = \frac{1}{t}\ln\frac{a}{a-x} = \frac{2.303}{t}\log\frac{a}{a-x} \quad ..(2)$$

If $[A_0]$ and [A] be the concentrations of reactant at zero time and time t, respectively then Eq. 2 may be put as

$$k = \frac{1}{t}\ln\frac{[A_0]}{[A]}$$

This is the integrated rate expression for first order reaction. Unit of the rate constant

$$k = \frac{1}{t}\ln\frac{a}{a-x}$$

$kt = \ln a - \ln(a-x)$

$a - x = ae^{-kt}$

$x = a(1 - e^{-kt})$

The differential rate expression for nth order reaction is as follows.

$$\frac{dx}{dt} = k(a-x)^n \text{ or } k = \frac{dx}{(a-x)^n dt} = \frac{\text{concentration}}{(\text{concentration})^n\text{ time}} = \text{conc.}^{1-n}.\text{time}^{-1}$$

If concentration be expressed in mole L^{-1} and time in minute, then

$k = (\text{mole } L^{-1})^{1-n}\text{ min}^{-1}$

For zero order reaction: n = 0 and hence k = mole L^{-1} min^{-1}

For 1st order reaction: n = 1 and hence

$k = (\text{mole } L^{-1})^0\text{ min}^{-1} = \text{min}^{-1}$

For 2nd order reaction: n = 2 and hence

$k = (\text{mole } L^{-1})^{-1}\text{ min}^{-1} = \text{mole}^{-1}\text{ l min}^{-1}$

The rate constant of a first order reaction has only time unit. It has no concentration unit. This means the numerical value of k for a first order reaction is independent of the unit in which concentration is expressed. If concentration unit is changed the numerical value of k for a first order reaction will not change. However, it would change with change in time unit. Say, k is 6.0×10^{-3} min then it may also be written as $1 \times 10^{-4}\ s^{-1}$ i.e. numerical value of k will decrease 60 times if time unit is changed from however to minute or from minute to second.

Half-time or half-life period of a first order reaction

The half-time of a reaction is defined as the time required to reduce the concentration of the reactant to half of its initial value. It is denoted by the symbol $t_{1/2}$. Thus,

When $x = \frac{a}{2}$, $t = t_{1/2}$

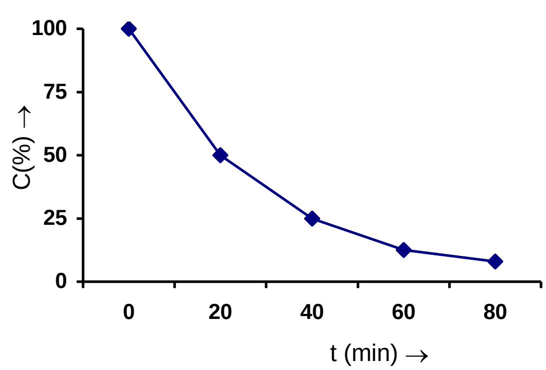
Putting these in equation 2 mentioned above, we get

$$k = \frac{2.303}{t_{1/2}} \log \frac{a}{a - \frac{a}{2}} = \frac{2.303}{t_{1/2}} \log 2 = \frac{2.303}{t_{1/2}} \times 0.30103$$

($\because$ log2 = 0.30103)

$$t_{1/2} = \frac{0.693}{k} \qquad \ldots(3)$$

Since k is a constant for a given reaction at a given temperature and the expression lacks any concentration term so from equation 3 it is evident that ***half-time of a 1st order reaction is a constant independent of initial concentration of reactant***. This means if we start with 4 mole L^{-1} of a reactant reacting by first-order kinetics and after 20 minute it is reduced to 2 mole L^{-1} will also be 20 minute. That is, after 20 minutes from the start of reaction the concentration of the reactant will be 2 mole L^{-1}, after 40 minutes from the start of reaction of concentration is 1 mole L^{-1}. After 60 minutes from the start of reaction the concentration of the reactant will be reduced to 0.5 mole L^{-1}. In other words, if during 20 minute 50% of the reaction completes, then in 40 minute 75%, in 60 minute 85.5% of the reaction and on will complete as shown with following plot.



Thus,

Fraction left after n half-lives = $\left(\frac{1}{2}\right)^n$

Concentration left after n half lives $a_n = \left(\frac{1}{2}\right)^n a_0$

It is also to be noted that equation 3 helps to calculate $t_{1/2}$ or k with the knowledge of k or $t_{1/2}$.

A general expression for $t_{1/2}$ is as follows

$$t_{1/2} \propto \frac{1}{a^{n-1}}$$

where n = order of reaction.

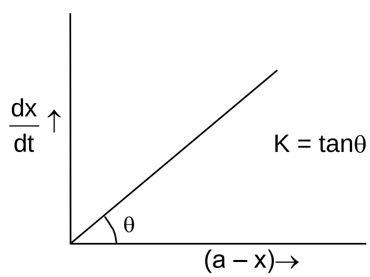
Graphical Representation

Since for nth order reaction

$$\frac{dx}{dt} = k(a-x)^n, \text{ or } -\frac{dx}{dt} = k.c^n$$

So, from this it is evident that a plot of $\frac{dx}{dt}$ vs $(a – x)^n$ will be a straight line passing through the origin and will have its slope equal to k, the rate constant of reaction.

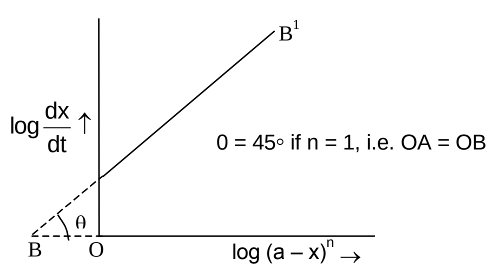
Thus, for a first order reaction one will get straight line passing through the origin of $\frac{dx}{dt}$ i.e. rate of reaction be plotted against a – x as shown below.



Taking logarithm of the above equation

$$\log\frac{dx}{dt} = n \log (a – x) + \log k$$

This equation demands that a plot of $\log\frac{dx}{dt}$ vs log (a – x) will be straight line of the slope equal to n, order of reaction and intercept equal to log k. For first order reaction this slope will be 1 as shown below.



Equation 2 may be rearranged as

$$\log (a - x) = \left(\frac{-k}{2.303}\right)t + \log a$$

Thus, a plot of log (a – x) vs. t will be straight line with slope equal to $-\frac{k}{2.303}$ and intercept equal to log a, if the reaction is of first order.

Exercise 1:

a) For first order reaction A $\longrightarrow$ Product, establish a relationship between rate constant and degree of dissociation of A.

b) For the reaction A + B $\longrightarrow$ C + D, the rate law equation is Rate = K_1[A] + K_2[A][B], where K_1 and K_2 represent two different constants. The products of this reaction are formed by two different mechanisms. What may be said about the relative magnitude of the rates of the two individual mechanisms?

4. Half – Life of a nth Order Reaction

Let us venture on to find out the $t_{½}$ for a nth order reaction where n ≠1.

$$\therefore\ \frac{-d[A]}{dt} = k[A]^n \Rightarrow \frac{-d[A]}{[A]^n} = kdt \Rightarrow -\int_{[A_o]}^{[A_o]/2} \frac{d[A]}{[A]^n} = k\int_0^{t_{1/2}} dt$$

$$\int_{[A_o]/2}^{[A_o]} [A]^{-n} d[A] = kt_{1/2} \quad \left[\frac{[A]^{1-n}}{1-n}\right]_{A_{0/2}}^{A_0} = kt_{1/2}$$

$$\Rightarrow \frac{1}{1-n}\left([A]_0^{1-n} - \left[\frac{[A]_0}{2}\right]^{1-n}\right) = kt_{1/2} \Rightarrow \frac{[A]_0^{1-n}}{1-n}\left[1-\left(\frac{1}{2}\right)^{1-n}\right] = kt_{1/2}$$

$$\Rightarrow \frac{1}{(1-n)[A_o]^{n-1}}\left[1-2^{n-1}\right] = kt_{1/2} \Rightarrow \frac{\left(2^{n-1}-1\right)}{(n-1)[A_o]^{n-1}} = kt_{1/2} \quad \textbf{(order n≠1)}$$

Therefore for a nth order reaction, the half life is inversely related to the initial concentration raised to the power of (n–1).

Note : It can be noted that for a zero order reaction $t_{1/2} = \frac{[A]_0}{2k}$.

Exercise 2:

Write a rate law expression for the overall reaction between A and B to yield C and D by the two mechanisms proposed below. What will be the initial rate of conversion of A and B in a solution containing 0.1 M of each? At what concentrations of A and /or B will the inherent rates be equal?

i) A + B $\longrightarrow$ AB* $\longrightarrow$ C + D ***$K_1' = 1 \times 10^{-5}\ m^{-1}s^{-1}$***

ii) $\begin{cases} A \longrightarrow A^* \longrightarrow E & K_1 = 1\times10^{-4}\ \sec^{-1} \\ E + B \longrightarrow C + D & K_2 = 1\times10^{10} M^{-1}s^{-1} \end{cases}$

Effect of Temperature on the Reaction Rate (Arrhenius Theory)

Temperature has very marked effect on the reaction rate. It has been found that the rate of most homogeneous reactions are nearly doubled or tripled by 10° rise in temperature.

$$\frac{k_{t+10}}{k_t} = 2 \text{ to } 3(\text{nearly})$$

The ratio of the rate constants of a reaction at two different temperatures differing by 10° i.e. k_{t+10}/k_t is known as temperature coefficient of reaction rate. This ratio also depends upon temperature and two temperatures generally selected are 25°C and 35°C. If a reaction has a temperature coefficient of reaction rate equal to 3, then by raising its temperature from 25°C to 65°C, the rate will increase by nearly $3 \times 3 \times 3 \times 3$ i.e. 81 times.

In order to explain the effect of temperature on the reaction rate. Arrhenius proposed a theory of reaction rate which states as follows :

i) A chemical reaction takes place by collision between the reactant molecules, and collision to be effective the colliding molecules must posses some certain minimum energy called threshold energy of the reaction.

ii) Reactant molecules having energy equal or greater than the threshold are called active molecules and those having energy less than the threshold are called passive molecules.

iii) At a given temperature there exists a dynamic equilibrium between active and passive molecules. The process of transformation from passive to active molecules being endothermic, increase of temperature increases the number of active molecules and hence the reaction.

   Passive molecules $\rightleftharpoons$ Active molecules, $\Delta H = +ve$

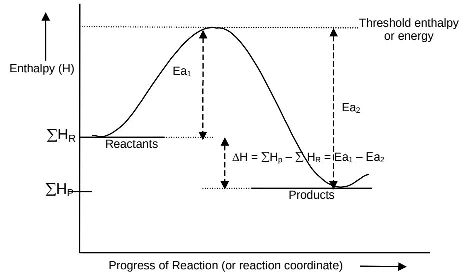
iv) Concept of energy of activation (E_a)

   The extra amount of energy which the reactant molecules (having energy less than the threshold) must acquire so that their mutual collision may lead to the breaking of bond(s) and hence the reaction, is known as energy of activation of the reaction. It is denoted by the symbol E_a. Thus,

   E_a = Threshold energy – Actual average energy

   E_a is expressed in kcals mole^{-1} or kJ mole^{-1}.

The essence of Arrhenius Theory of reaction rate is that there exists an energy barrier in the reaction path between reactant(s) and product(s) and for reaction to occur the reactant molecules must climb over the top of the barrier which they do by collision. The existence of energy barrier and concept of E_a can be understood from the following diagram.



$\sum H_R$ =Summation of enthalpies of reactants

$\sum H_P$ =Summation of enthalpies of products

ΔH = Enthalpy change during the reaction

Ea_1= Energy of activation of the forward reaction (FR)

Ea_2= Energy of activation of the backward reaction (BR)

Notable Points

i) ΔH = Energy activation of FR – Energy of activation of BR

ii) In the diagram mentioned above, FR is exothermic (ΔH is –ve) while BR is endothermic (ΔH is +ve). The minimum activation energy of FR i.e. any exothermic reaction will be zero while minimum activation energy for BR i.e. any endothermic reaction will be equal to ΔH.

iii) Greater the height of energy barrier, greater will be the energy of activation and more slower will be the reaction at a given temperature.

iv) Rate = Collision frequency $\times$ fraction of the total number of collision which is effective.

or

= Collision frequency $\times$ fraction of the total number of collisions in which K.E. of the colliding molecules equals to E_a or exceeds over it.

Collision frequency is the number of collisions per unit volume per unit time. It is denoted by the symbol Z. Z is directly proportional to $\sqrt{T}$. By 10° rise in temperature. So, it is the fraction of the total number of effective, collisions that increases markedly by 10° rise in temperature resulting into marked increase in the reaction rate.

Arrhenius Equation

The variation equilibrium constant of a reaction with temperature is described by Van’t Hoff equation of thermodynamics which is as follows:

$$\frac{d\ln K_p}{dT} = \frac{\Delta H}{RT^2}$$

If k_1 and k_2 be the rate constants of FR and BR, respectively then $K_p = k_1/k_2$. Further, $\Delta H = Ea_1 - Ea_2$ .Putting these in the above equation we get,

$$\frac{d\ln k_1}{dT} - \frac{d\ln k_2}{dT} = \frac{E_{a1}}{RT^2} - \frac{E_{a2}}{RT^2}$$

Splitting into two parts

$$\frac{d\ln k_1}{dT} = \frac{E_{a1}}{RT^2} + Z \quad \text{(For FR)}$$

$$\frac{d\ln k_2}{dT} = \frac{E_{a_2}}{RT^2} + Z \text{ (For BR)}$$

where Z is constant

Arrhenius sets Z equal to are and without specifying FR and BR, he gave the following equation called Arrhenius equation.

$$\frac{d\ln k}{dT} = \frac{E_a}{RT^2} \qquad \text{...(4)}$$

From this equation it is evident that rate of change of logarithm of rate constant with temperature depends upon the magnitude of energy of activation of the reaction. Higher the Ea_1 smaller the rate of change of logarithm rate constant with temperature. That is, rate of the reaction with low E_a increases slowly with temperature while rate of the reaction with high E_a increases rapidly with temperature. It is also evident that rate of increase of logarithm of rate constant will go on decreasing with increase of temperature.

Integrating Equation 4 assuming Ea to be constant we get,

$$\ln k = -\frac{Ea}{RT} + \ln A \qquad \text{...(5)}$$

$$\text{or} \quad \ln\frac{k}{A} = -\frac{Ea}{RT}$$

$$\text{or} \quad k = Ae^{-Ea/RT} \qquad \text{...(6)}$$

Equation (6) is integrated form of Arrhenius equation. The constant A called pre-exponential factor is the frequency factor since it is somewhat related with collision frequency. It is a constant for a given reaction. From Equation 6 it is evident that as $T \to \infty$, $k \to A$. Thus, the constant A is the rate constant of reaction at infinity temperature. The rate constant goes on increasing with temperature.

So, when T approaches infinity, k will be maximum. That is to say, A is the maximum rate constant of a reaction.

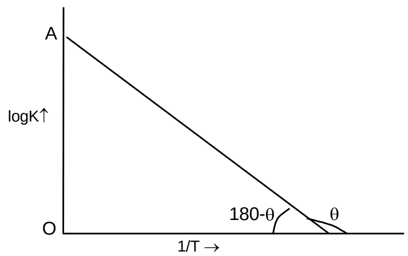
It is also to be noted that the exponential term i.e. $e^{-Ea/RT}$ measures the fraction of total number of molecules in the activated state or fraction of the total number of effective collisions. If n_{Ea} and n be the number of molecules of reactant in the activated state and the total number of molecules of the reactant present in the reaction vessel respectively, then

$$\frac{n_{E_a}}{n} = e^{-E_a/RT}$$

Equation 5 may also be put as

$$\log k = \left(-\frac{Ea}{2.303R}\right)\frac{1}{T} + \log A \qquad \text{...(7)}$$

Since $\frac{E_a}{2.303R}$ and logA both are constants for a given reaction. So from equation 7 it is evident that a plot of log k vs. $\frac{1}{T}$ will be a straight line of the slope equal to $-\frac{E_a}{2.303R}$ and intercept equal to logA as shown below.



$$\frac{-E_a}{2.303R} = \tan\theta = -\tan(180-\theta) = -\frac{0A}{OB}$$

$$\therefore\ E_a = \frac{OA}{OB} \times 2.303R$$

$$\log A = OA$$

Thus, from this plot E_a and A both can be determined accurately.

If k_1 be the rate constant of a reaction at two different temperature T_1 and T_2 respectively then from equation 7, me may write

$$\log k_1 = -\frac{E_a}{2.303R} \cdot \frac{1}{T_2} + \log A$$

$$\log k_2 = -\frac{E_a}{2.303R} \cdot \frac{1}{T_2} + \log A$$

Subtracting former from the latter we get

$$\log\frac{k_1}{k_1} = \frac{E_a}{2.303R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right) \qquad \ldots(8)$$

With the help of this equation it is possible to calculate E_a of a reaction provided, rate constants of reaction at two different temperatures are known. Alternatively one can calculate rate constant of a reaction at a given temperature provided that rate constant of the reaction at some other temperature and also E_a of the reaction is known.

Exercise 3:

a) ***In a certain polluted atmosphere containing O_3 at a steady state concentration of 2.0×10^{-8} mol/L, the hourly production of O_3 by all sources was estimated as 7.2×10^{-15} mol/L. If the only mechanism for destruction of O_3 is the second order reaction $2O_3 \longrightarrow 3O_2$, calculate the rate constant for the destruction reaction, defined by the rate law for $-\Delta[O_3]/\Delta t$.***

b) ***From the following data, estimate the activation energy for the reaction $H_2 + I_2 \longrightarrow 2HI$***

T, K	$\frac{1}{T}, K^{-1}$	***logK***
769	1.3×10^{-3}	***2.9***
667	1.5×10^{-3}	***1.1***

5. Catalyst

A catalyst is a substance, which increases the rate of a reaction without itself being consumed at the end of the reaction, and the phenomenon is called catalysis. There are some catalysts which decrease the rate of reaction and such catalysts are called negative catalyst. Obviously, the catalyst accelerating the rate will be positive catalyst. However, the term positive is seldom used and catalyst itself implies positive catalyst.

Catalyst are generally foreign substances but sometimes one of the product formed may act as a catalyst and such catalyst is called ***"auto catalyst"*** and the phenomenon is called auto catalysis. Thermal decomposition of $KClO_3$ is found to be accelerated by the presence of MnO_2. Here MnO_2 (foreign substance) acts as a catalyst.

$$2KClO_3 + [MnO_2] \longrightarrow 2KCl + 3O_2\uparrow + [MnO_2]$$

MnO_2 can be received in the same composition and mass at the end of the reaction. In the permanganate titration of oxalic acid in the presence of bench H_2SO_4 (acid medium), it is found that the titration in the beginning there is slow discharge of the colour of permanganate solution but after sometime the discharge of the colour become faster. This is due to the formation of $MnSO_4$ during the reaction which acts as a catalyst for the same reaction. Thus, $MnSO_4$ is an ***"auto catalyst"*** for this reaction. This is an example of auto catalyst.

$$2KMnO_4 + 3H_2SO_4 + 5H_2C_2O_2 \longrightarrow K_2SO_4 + 8H_2O + 10CO_2$$

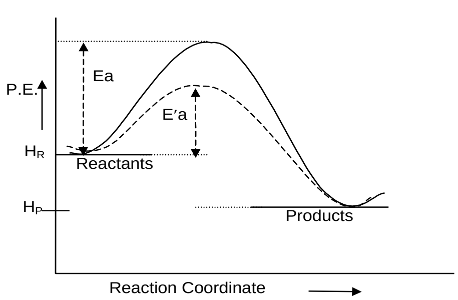
General characteristics of catalyst

i) A catalyst does not initiate the reaction. It simply fastens it.

ii) Only a small amount of catalyst can catalyse the reaction.

iii) A catalyst does not alter the position of equilibrium i.e. magnitude of equilibrium constant and hence ΔG^0. It simply lowers the time needed to attain equilibrium. This means if a reversible reaction in absence of catalyst completes to go to the extent of 75% till attainment of equilibrium, and this state of equilibrium is attained in 20 minutes then in presence of a catalyst also the reaction will go to 75% of completion before the attainment of equilibrium but the time needed for this will be less than 20 minutes.

iv) A catalyst drives the reaction through a different route for which energy barrier is of shortest height and hence E_a is of lower magnitude. That is, the function of the catalyst is to lower down the activation.



E_a = Energy of activation in absence of catalyst.
$E'a$ = Energy of activation in presence of catalyst.
$Ea - E'_a$ = lowering of activation energy by catalyst.

If k and k_{cat} be the rate constant of a reaction at a given temperature T, and E_a and E'_a are the activation energies of the reaction in absence and presence of catalyst, respectively, the

$$\frac{k_{cat}}{k} = \frac{Ae^{-E'a/RT}}{Ae^{-E_a/RT}}$$

$$\frac{k_{cat}}{k} = Ae^{(E_a - E'_a)/RT}$$

Since E_a, E_a' is + so $k_{cat} > k$. the ratio $\frac{k_{cat}}{k}$ gives the number of times the rate of reaction will increase by the use of catalyst at a given temperature and this depends upon E_a - E_a^1. Greater the value of $E_a – E_a^1$, more number of times k_{cat} is greater than k.

The rate of reaction in the presence of catalyst at any temperature T_1 may be made equal to the rate of reaction in absence of catalyst but for this sake we will have to raise the temperature. Let this temperature be T_2 this

$$e^{-E'_a/RT_1} = e^{-E_a/RT_2}$$

$$\text{or } \frac{E'_a}{T_1} = \frac{E_a}{T_2} \qquad \ldots(9)$$

Studying the progress of reaction

In order to follow the progress of reaction, the concentration of any of the reactants or products which ever is convenient may be determined at various time intervals using some suitable chemical method. At each time interval the progress of the reaction is arrested by immersing the reaction vessel in freezing mixture. The temperature of the freezing mixture being very low (less than 0°C), the rate of reaction is reduced to almost nothing. It is not necessary to determine always the concentration but one can determine any other parameter that changes during the reaction with time and which directly proportional to the concentration of the reactant or product. These facts are illustrated below by some examples.

Illustration 1: ***The progress of this reaction given below can be followed by measuring the concentration of acid (HCl acid used as catalyst plus acetic acid formed during the reaction) by means of alkali titration. Find the volume of alkali NaOH needed for the end point that will increase with time.***

Solution:

$$CH_3COOC_2H_5 + H_2O \xrightarrow{H^+} CH_3COOH + C_2H_5OH$$

a	excess	0	0	… t = 0
a – x	constant	x	x	… t = t
0		a	a	… t = ∞

If V_0, V_t and V_∞ are the volumes of NaOH solution needed for the end point of titration of the reaction mixture at zero time, time t and at infinity i.e. after completion of the reaction — the condition being achieved by heating the reaction mixture for some time, respectively then

$V_0 \propto$ [acid catalyst]
$V_t \propto$ [acid catalyst] + x
$V_\infty \propto$ [acid catalyst] + a
$\therefore V_\infty - V_t \propto a - x$
$V_\infty - V_0 \propto a$
(since concentration of HCl acid acting as catalyst will remain constant).

The above reaction which is of first order (actually pseudo unimolecular) will, therefore, obey following equation.

$$k = \frac{2.303}{t}\log\frac{V_\infty - V_0}{V_\infty - V_t}$$

Example 1: $H_2O_2\,(aq) \longrightarrow H_2O + \frac{1}{2}O_2\uparrow$

a	0	0	… t = 0
a – x	x	x	… t = t

Since H_2O_2 acts as a reducing agent towards $KMnO_4$, so concentrations of H_2O_2 at various time intervals may be determined by the titration of the reaction mixture against standard $KMnO_4$ solution. The titre value will go on decreasing with time.

If V_0 and V_t be the titre values at zero time and any time t then

$V_0 \propto a$ and $V_t \propto a - x$

The above reaction being first-order, its rate constant may be expressed as

$$k - \frac{2.303}{t}\log\frac{V_0}{V_t}$$

Example 2: The reaction mentioned below is first-order w.r.t. sucrose and zero order w.r.t. water since water is in large excess as compared to sucrose. That is, it is an example of pseudo unimolecular reaction. Sucrose, glucose and fructose all are optically active substance. So, the progress of the reaction can be followed by measuring angle of rotations of the reaction mixture at various time intervals.

During the reaction angle of rotation goes on decreasing and after sometime there is reversal of the direction of rotation i.e. from dextro to laevo and hence the reaction is called ***"inversion of cane sugar"*** or inversion of sucrose.

$$\underset{d-\text{Sucrose}}{C_{12}H_{22}O_{11}} + H_2O \xrightarrow{H^+} \underset{d-\text{Glucose}}{C_6H_{12}O_6} + \underset{l-\text{Furctose}}{C_6H_{12}O_6}$$

Initially	a	Excess	0	0
After time t	a – x	constant	x	x
At infinity	0	constant	a	a

Angle of optical rotation is measured by means of an instrument called polarimeter. Optical rotation is mathematically expressed as

$R_{obs} = l.C.\ [\propto]_D^t$

Where l = length of the polarimeter tube

C = concentration of test solution

$[\propto]_D^t$ = specific rotation

For a given sample and polarimeter, l and $[\propto]_D^t$

$r_{obs} \propto C$, or $r_{obs} = kC$, k – proportionality constant

If r_0, r_t and r_∞ be the observed angle of rotations of the sample at zero time, time t and infinity respectively, and k_1, k_2 and k_3 be the proportionately in terms of sucrose, glucose and fructose, respecting.

Then,

$r_0 = k_1a$

$r_t = k_1(a - x) + k_2x + k_3x$

$r_\infty = k_2a + k_3a$

From these equations it can be shown that

$$\frac{a}{a-x} = \frac{r_0 - r_\infty}{r_t - r_\infty}$$

So, the expression for the rate constant of this reaction in terms of the optical rotational data may be put as

$$k = \frac{2.303}{t}\log\frac{r_0 - r_\infty}{r_t - r_\infty}$$

Example 3:

$2N_2O_5 \longrightarrow 4NO_2 + O_2$

P_0	0	0	… t = 0
$P_0 - 2x$	4x	x	... t = t
0	$2P_0$	½ P_0	... $t = \infty$

The progress of the reaction can be followed by measuring the pressure of the gaseous mixture in a closed vessel i.e. at constant volume. The expression for the rate constant in terms of pressure data will be as given below.

$$k = \frac{2.303}{t}\log\frac{P_0}{P_t}$$, when $P_t = P_0 - 2x$

Total pressure after any time t and at ∞ be given it is possible to find P_0 and x and hence k may be calculated.

6. Some Simple First Order Reactions

As we have to generally deal with only first order reactions, we examine some of these reactions from the point of calculating the rate constant based on different experimental data. Now we present several problems in which we shall learn how to calculate the rate constant of reactions based on the variety of data given.

We are given a first order reaction

$A \rightarrow B+C$ where we assume that A,B and C are gases. The data given to us is

Time	0	t
Partial pressure of A	P_1	P_2

And we have to find the rate constant of the reaction.

Since A is a gas and assuming it to be ideal, we can state that $P_A = [A]RT$ [From PV = nRT]. $\therefore$ at t = 0, $P_1 = [A]_o RT$ and at t = t, $P_2 = [A]_t RT$. Thus the ratio of the concentration of A at two different time intervals is equal to the ratio of its partial pressure at those same time intervals.

$$\therefore \frac{[A]_o}{[A]_t} = \frac{P_1}{P_2}$$

$$\therefore \mathbf{k} = \frac{\mathbf{1}}{\mathbf{t}} \mathbf{ln} \frac{\mathbf{P_1}}{\mathbf{P_2}}$$

Now we consider the same reaction with different set of data

$A \longrightarrow B + C$

Time	0	t
Total pressure of A +B+C	P_1	P_2

Find k.

In this we are given total pressure of the system at these time intervals. The total pressure obviously includes the pressure of A, B and C. At t = 0, the system would only have A. Therefore the total pressure at t = 0 would be the initial pressure of A $\therefore$ P_1 is the initial pressure of A. At time t let us assume that some moles of A decomposed to give B and C because of which its pressure is reduced by an amount x while that of B and C is increased by x each. That is

	A $\longrightarrow$	B +	C
Initial	P_1	0	0
At time t	P_1–x	x	x

$\therefore$ total pressure at time t = $P_1 + x = P_2$

$\Rightarrow x = P_2 - P_1$

Now the pressure of A at time t would be $P_1 - x = P_1 - (P_2 - P_1) = 2P_1 - P_2$

$$\therefore k = \ln \frac{[A]_0}{[A]_t} = \mathbf{ln} \frac{\mathbf{P_1}}{(\mathbf{2P_1 - P_2})}$$

Now let us move on to the next case. In this case we have $A \longrightarrow B + C$

Time	t	∞
Total pressure of A+B+C	P_2	P_3

Find k.

Here ∞ means that the reaction is complete. Now we have

	A $\longrightarrow$	B +	C
At t=0	P_1	0	0
At t =t	$(P_1 - x)$	x	x
At t =∞	0	P_1	P_1

$\therefore 2P_1 = P_3$

$\Rightarrow P_1 = \frac{P_3}{2}$

At time t,

$P_1 + x = P_2$

$\Rightarrow \frac{P_3}{2} + x = P_2 \Rightarrow x = P_2 - \frac{P_3}{2}$

$\Rightarrow P_1 - x = \frac{P_3}{2} - (P_2 - \frac{P_3}{2}) = P_3 - P_2$

$$\therefore\ k = \frac{1}{t}\ln\frac{[A]_0}{[A]_t} = \frac{1}{t}\ln\frac{P_3/2}{(P_3 - P_2)} = \mathbf{\frac{1}{t}\ln\frac{P_3}{2(P_3 - P_2)}}$$

Illustration 2: ***A $\longrightarrow$ B +C***

Time	***t***	***∞***
Total pressure of (B+C)	***P_2***	***P_3***

Find k.

Solution:

	A $\longrightarrow$	B +	C
At t=0	P_1	0	0
At t =t	$P_1 - x$	x	x
At t = ∞	0	P_1	P_1

$\therefore 2P_1 = P_3$

$\Rightarrow P_1 = \frac{P_3}{2}$

$2x = P_2 \Rightarrow x = \frac{P_2}{2}$

$\therefore P_1 - x = \frac{P_3}{2} - \frac{P_2}{2} = \frac{P_3 - P_2}{2}$

$$\therefore\ k = \frac{1}{t}\ln\frac{[A]_0}{[A]_t} = \mathbf{\frac{1}{t}\ln\frac{P_3}{(P_3 - P_2)}}$$

Exercise 4: ***A$\longrightarrow$B + C***

Time	***t***	***∞***
Partial pressure of B	***P_2***	***P_3***

Find k.

Now let us assume that A,B and C are substances present in a solution. From a solution of A, a certain amount of the solution (small amount) is taken and titrated with a suitable reagent that reacts with A. The volume of the reagent used is V_1 at t = 0 and V_2 at t = t

Time	0	T
Volume of reagent	V_1	V_2

The reagent reacts only with A. Find k.

It can be understood that the volume of the reagent consumed is directly proportional to the concentration of A. Therefore the ratio of volume of the reagent consumed against A at t=0 and t = t is equal to $[A]_0/[A]_t$

$$\therefore\ k = \frac{1}{t}\ln\frac{[A]_0}{[A]_t} = \frac{1}{t}\ln\frac{V_1}{V_2}$$

Illustration 3: ***If $A \longrightarrow B + C$***

Time	***0***	***t***
Volume of reagent	***V_1***	***V_2***

The reagent reacts with A,B and C. Find k.

Solution: As we had seen in the previous case that the result of illustration 5 matches that of illustration of 1, we can use a similar logic for solving this one. That is

	A $\longrightarrow$	B +	C
At t = 0	V_1	0	0
At t = t	$V_1 - x$	x	x
At t = ∞	0	V_1	V_1

where x is the volume of the reagent for those moles of A which have been converted into B and C

$\therefore\ V_1 + x = V_2$

$\Rightarrow x = V_2 - V_1$

$\Rightarrow V_1 - x = 2V_1 - V_2$

$$\therefore\ \mathbf{k} = \frac{1}{t}\ln\frac{[A]_0}{[A]_t} = \frac{1}{t}\ln\frac{V_1}{(2V_1 - V_2)}$$

This result looks perfectly OK. The problem is that this is true only if we make an assumption in the beginning of this problem (which you may have thought about). The assumption is that the 'n' factor of A,B and C with the reagent is same.

This can be made clear as follows. Let y moles of A be converted to B and C. If y moles of A require x ml of reagent and if the 'n' factor of A is n_1 with the reagent then the equivalents of A converted are n_1y and in x ml of reagent n_1y equivalents of the reagent are present. Now since the same y moles of B and C are formed and we find the same volume of reagent used for B and C, it can be calculated that y moles of B and C also contain n_1y equivalents, each, which implies that 'n' factor of B and C are also 'n_1'.

Exercise 5: ***$A \longrightarrow B + C$***

Time	***0***	***t***
Volume of reagent	***V_1***	***V_2***

Reagent reacts with all of them (A, B & C). A,B and C have 'n' factors in the ratio of 1:2:3 with the reagent (assuming its n factor remains same with all of them). Find k.

Now consider this situation $A \longrightarrow B+C$

Time	T	∞
Volume of reagent	V_2	V_3

Reagent reacts with all A,B and C and have 'n' factors in the ratio of 1:2:3 with the reagent. Find k.

	A	$\longrightarrow$ B	+ C
At t = 0	V_1	0	0
At t = t	$V_1 - x$	2 x	3x
At t = ∞	0	$2V_1$	$3V_1$

$\therefore 5V_1 = V_3$

$V_1 = V_3/5$

$V_1 + 4x = V_2$

$$x = \frac{V_2 - V_1}{4}$$

$$\therefore V_1 - x = \frac{V_3}{5} - \frac{V_2}{4} + \frac{V_3}{20} = \frac{5V_3 - 5V_2}{20}$$

$$\therefore k = \frac{1}{t}\ln\frac{[A]_0}{[A]_t} = \frac{1}{t}\ln\frac{V_3/5}{5V_3 - 5V_2/20}$$

$$k = \frac{\mathbf{1}}{\mathbf{t}}\mathbf{ln}\frac{\mathbf{4V_3}}{\mathbf{5(V_3 - V_2)}}$$

Now, we will consider a reaction $A \xrightarrow{D} B+C$ which is catalysed by D. We will assume in this problem that the concentration of the catalyst remains constant throughout. The data given to us is

Time	**0**	**t**	∞
Volume of reagent	V_1	V_2	V_3

The reagent reacts with all (A,B,C and D). Assume that 'n' factor of A,B and C are in the ratio of 1:1:1 and that of D is not known. Find k.

Let V_A be the volume of the reagent required by A initially and V_D be the volume required by D.

$\therefore V_1 = V_A + V_D$

$V_2 = (V_A - x) + x + x + V_D$ (x is the volume of the reagent required for those moles of A that have reacted to give B and C).

$V_3 = 2V_A + V_D$

We can see that, $V_3 – V_1 = V_A$

And $V_3 – V_2 = V_A – x$

$$\therefore\ \mathbf{k} = \frac{\mathbf{1}}{\mathbf{t}}\ln\frac{[\mathbf{A}]_0}{[\mathbf{A}]_t} = \frac{\mathbf{1}}{\mathbf{t}}\ln\frac{(\mathbf{V_3} - \mathbf{V_1})}{(\mathbf{V_3} - \mathbf{V_2})}$$

Exercise 6: $A \xrightarrow{D} B+C$

Time	*0*	*t*	*∞*
Volume of reagent	V_1	V_2	V_3

The reagent reacts with only B,C and D. Find k.

If $A \xrightarrow{D} B+C$

Time	0	t	∞
Volume of reagent	V_1	V_2	V_3

And the reagent reacts with only C and D. Find k.

$V_1 = V_D$

$V_2 = x + V_D$

$V_3 = V_A + V_D$

$V_3 – V_1 = V_A$

$V_3 – V_2 = V_A – x$

$$\therefore k = \frac{1}{t}\ln\frac{[A]_0}{[A]_t} = \frac{\mathbf{1}}{\mathbf{t}}\ln\frac{(\mathbf{V_3} - \mathbf{V_1})}{(\mathbf{V_3} - \mathbf{V_2})}$$

Exercise 7: $A \xrightarrow{D} B+C$

Time	**0**	**t**	**∞**
Volume of reagent	V_1	V_2	V_3

The reagent reacts with all of them. The ‘n’ factor of A,B and C are in the ratio of 1:2:3 with the reagent. Find k.

Now we shall see how to find the rate constant of a reaction using a very different set of data. There are some organic compounds which have a property of rotating a plane polarized light in a particular direction by a particular value. The compounds are called optically active compounds. One reaction in which an optically active substance converts to some other optically active substance is,

$$\text{Sucrose} \xrightarrow{H^+} \text{Glucose} + \text{Fructose}$$

Sucrose, Glucose and Fructose are all optically active and while the first two compounds are dextro rotatory (rotating the plane polarised light in the right hand direction and the last is laevo rotatory (rotating the plane polarized light in the left hand direction). All the three compounds rotate the plane polarised light by different angles and their rotation is directly proportional to concentration.

Now the problem is $S \longrightarrow G + F$ and the data is

Time	0	t
Rotation of sucrose	r_0	r_t

Find k.

Let the rotation of Sucrose be $r_1°$ per mole and the initial moles of Sucrose be a.

$\therefore r_0 = ar_1^o$

Let the moles of Sucrose that is converted to Glucose and Fructose be x.

$\therefore r_t = (a–x)\ r_1^o$

$\therefore \dfrac{a}{a-x} = \dfrac{r_0}{r_t}$

$\therefore k = \mathbf{\dfrac{1}{t}\ln\dfrac{a}{a-x} = \dfrac{1}{t}\ln\dfrac{r_0}{r_t}}$

Exercise 8: ***a)*** $S \longrightarrow G + F$

Time	*t*	∞
Rotation of Glucose & Fructose	r_t	R_∞

Find k.

b) $S \longrightarrow G + F$

Time	*0*	*t*	∞
Rotation	R_o	r_t	R_∞

The rotation is due to all (total rotation). Find k.

Now let us assume that the first order reaction is $A \longrightarrow B+C$ and this solution is taken in a tube. We radiate this solution with a monochromatic light of wavelength λ and we will assume that this radiation is absorbed only by A. Let the intensity of incident light be I_0 and that of the transmitted light be I_t. $\log\dfrac{I_t}{I_o}$ is called transmittance and it is inversely proportional to the concentration of A.

That is $\log\dfrac{I_t}{I_o}$ = transmittance $\propto \dfrac{1}{[A]}$

The data given is

Time	0	t
$\log I_t/I_0$	x	y

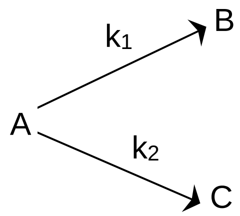
Find k.

$x \propto \dfrac{1}{[A]_0}$ and $y \propto \dfrac{1}{[A]_t}$

$\therefore \mathbf{k} = \mathbf{\dfrac{1}{t}\ln\dfrac{[A]_0}{[A]_t} = \dfrac{1}{t}\ln\dfrac{y}{x}}$

7. Some Complex First Order Reactions

7.1 Parallel Reactions



In such reactions (mostly organic) a single reactant gives two products B and C with different rate constants. If we assume that both of them are first order, we get.

$$\frac{-d[A]}{dt} = k_1[A] + k_2[A] = (k_1 + k_2)[A] \qquad \ldots(1)$$

$$\frac{d[B]}{dt} = k_1[A] \qquad \ldots(2)$$

and $\frac{d[C]}{dt} = k_2[A]$ …(3)

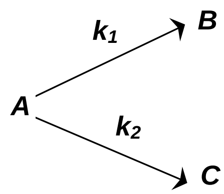
Let us assume that in a time interval, dt, x moles / lit of B was produced and y moles / lit of C was produced.

$$\therefore \frac{d[B]}{dt} = \frac{x}{dt} \text{ and } \frac{d[C]}{dt} = \frac{y}{dt}$$

$\therefore \dfrac{\frac{d[B]}{dt}}{\frac{d[C]}{dt}} = \dfrac{x}{y}$. We can also see that from (2) and (3), $\dfrac{\frac{d[B]}{dt}}{\frac{d[C]}{dt}} = \dfrac{k_1}{k_2}$

$\therefore \dfrac{x}{y} = \dfrac{k_1}{k_2}$. This means that irrespective of how much time is elapsed, the ratio of concentration of B to that of C from the start (assuming no B and C in the beginning) is a constant equal to k_1/k_2.

Illustration 4:



Let $k_1:k_2 = 1:10$. Calculate the ratio, $\frac{[C]}{[A]}$ at the end of one hour assuming that $k_1 = x\ hr^{-1}$

Solution:

$$\frac{-d[A]}{dt} = (k_1 + k_2)[A]$$

$$\therefore \frac{-d[A]}{[A]} = (k_1+k_2)\, dt$$

Integrating with in the required limits, we get

$$\ln\frac{[A]_0}{[A]_t} = (k_1 + k_2)t$$

$$\therefore \ln\frac{[A]_t + [B] + [C]}{[A]_t} = (k_1+k_2)\,t$$

Since $\frac{[B]}{[C]} = \frac{k_1}{k_2} = \frac{1}{10}$

$$\therefore \ln\frac{[A]_t + \frac{[C]}{10} + [C]}{[A]_t} = 11x$$

$$\mathbf{\frac{[C]}{[A]_t} = \frac{10}{11}\left(e^{11x} - 1\right)}$$

7.2 Sequential Reactions

$A \xrightarrow{k_1} B \xrightarrow{k_2} C$. In this A decomposes to B which in turn decomposes to C.

$\therefore \frac{-d[A]}{dt} = k_1[A]$ …(1)

$\frac{d[B]}{dt} = k_1[A] - k_2[B]$ …(2)

$\frac{d[C]}{dt} = k_2[B]$ …(3)

Integrating equation (1), we get

$[A] = [A]_0 e^{-k_1 t}$

Now we shall integrate equation (2) and find the concentration of B related to time t.

$$\frac{d[B]}{dt} = k_1[A] - k_2[B] \Rightarrow \frac{d[B]}{dt} + k_2[B] = k_1[A]$$

substituting [A] as $[A]_0 e^{-k_1 t}$

$\Rightarrow \frac{d[B]}{dt} + k_2[B] = k_1[A]_0 e^{-k_1 t}$ …(4)

Integration of the above equation is not possible as we are not able to separate the two variables, [B] and t. Therefore we multiply equation (4) by an integrating factor $e^{k_2 t}$, on both the sides of the equation.

$$\left(\frac{d[B]}{dt} + k_2[B]\right)e^{k_2 t} = k_1[A]_0\, e^{(k_2 - k_1)t}$$

We can see that the left hand side of the equation is a differential of $[B]e^{k_2 t}$.

$$\therefore \frac{d}{dt}\left([B]e^{k_2 t}\right) = k_1[A]_0 e^{(k_2 - k_1)t}$$

$$d\left([B]e^{k_2t}\right) = k_1[A]_0 e^{(k_2-k_1)t}\,dt$$

Integrating with in the limits 0 to t.

$$\int d\left([B]e^{k_2t}\right) = k_1[A]_o \int_o^t e^{(k_2-k_1)t}\,dt \Rightarrow [B]e^{k_2t} = k_1[A]_0 \left[\frac{e^{(k_2-k_1)t}}{(k_2-k_1)}\right]_o^t$$

$$\Rightarrow [B]e^{k_2t} = \frac{k_1[A]_0}{(k_2-k_1)}\left[e^{(k_2-k_1)t}-1\right] \Rightarrow [B] = \frac{k_1[A]_0}{(k_2-k_1)}e^{-k_2t}\left[e^{(k_2-k_1)t}-1\right]$$

$$\textbf{[B]} = \frac{k_1[A]_0}{k_2-k_1}\left[e^{-k_1t}-e^{-k_2t}\right] \qquad \ldots(5)$$

Now in order to find [C], substitute equation (5) in equation (3), we get

$$\frac{d[C]}{dt} = \frac{k_1k_2[A]_0}{k_2-k_1}\left[e^{-k_1t}-e^{-k_2t}\right]$$

$$\therefore\ d[C] = \frac{k_1k_2[A]_0}{k_2-k_1}\left[e^{-k_1t}-e^{-k_2t}\right]dt$$

On integrating

$$\int d[C] = \frac{k_1k_2[A]_0}{(k_2-k_1)}\int_0^t\left[e^{-k_1t}-e^{-k_2t}\right]dt$$

$$\Rightarrow [C] = \frac{k_1k_2[A]_0}{(k_2-k_1)}\left[\left(\frac{e^{-k_1t}}{-k_1}\right)_0^t - \left(\frac{e^{-k_2t}}{-k_2}\right)_0^t\right] \Rightarrow [C] = \frac{k_1k_2}{(k_2-k_1)}[A]_0\left[\left(\frac{e^{-k_1t}-1}{-k_1}\right)-\left(\frac{e^{-k_2t}-1}{-k_2}\right)\right]$$

$$\Rightarrow [C] = \frac{k_1k_2}{k_2-k_1}[A]_o\left[\left(\frac{1-e^{-k_1t}}{k_1}\right)-\left(\frac{1-e^{-k_2t}}{k_2}\right)\right]$$

$$\textbf{[C]} = \frac{[A]_0}{k_2-k_1}\left[k_2\left(1-e^{-k_1t}\right)-k_1\left(1-e^{-k_2t}\right)\right]$$

B_{max} and t_{max}: We can also attempt to find the time when [B] becomes maximum. For this we differentiate equation (5) and find $\frac{d[B]}{dt}$ & equate it to zero.

$$\therefore\ \frac{d[B]}{dt} = \frac{k_1[A]_0}{(k_2-k_1)}\left[e^{-k_1t}(-k_1)+e^{-k_2t}(k_2)\right] = 0$$

$$\Rightarrow k_1e^{-k_1t} = k_2e^{-k_2t}$$

$\frac{k_1}{k_2} = e^{(k_1-k_2)t}$, taking log of both the sides

$$\therefore t_{max} = \frac{1}{k_1-k_2}\ln\frac{k_1}{k_2} \qquad \ldots(6)$$

Substituting equation (6) in equation (5)

$$\textbf{B}_{max} = [A]_o\left[\frac{k_2}{k_1}\right]^{k_2/k_1-k_2}$$

8. Radioactivity

All radioactive decay follow 1st order kinetics and this is where the similarity ends. This will be explained in a short while.

We measured the rate of reaction in chemical kinetics based on the rate of change of concentration of reactants or products. But this procedure will not work for calculating the rate of radioactive reaction. This is because most of the time the radioactive substance is a solid. Therefore its concentration would be a constant with time (assuming it to be pure and that the product does not remain with the reactants). Therefore the rate of radioactive reactions are measured by calculating the rate of change of nuclei of the radioactive substance. For a radioactive decay A $\rightarrow$B, the rate of reaction is calculated as

$$\frac{-dN_A}{dt} = \lambda\ N_A$$

Where λ = decay constant of reaction

N_A = number of nuclei of the radioactive substance at the time when rate is calculated.

As you can see the above rate law is very much similar to the rate law of a first order chemical reaction, but all other similarities ceases here. For example unlike a chemical reaction the decay constant (λ) does not depend on temperature. Arrhenius equation is not valid for radioactive decay.

$$\frac{-d_{N_A}}{dt} = \lambda N_A$$

Integrating the differential rate law we get

$$-\int_{N_o}^{N_t} \frac{d_{N_A}}{N_A} = \lambda \int_o^t dt$$

$log\,\frac{N_o}{N_t}$ = λ t Where N_o = number of nuclei of A at t = 0

N_t = number of nuclei of A at t = t

λ = decay constant

The expression can be rearranged to give

$N_t = N_o e^{-\lambda t}$ ... (1)

This suggests that the number of nuclei of radioactive substance A at any instant of time can be calculated, by knowing the number of nuclei at t = 0, its decay constant and the time.

8.1 Half – Life

Just like a 1st order reaction the half life of radioactive decay is given by

$$t_{1/2} = \frac{0.693}{\lambda}$$

Note: Let us start with 10 nuclei. If the half life is 5 minutes, then at the end of first 5 minutes, number of nuclei would be 5. Now what would be the number of nuclei after next 5 minutes? Will it be 2.5 or 2 or 3? We can clearly see that it cannot be 2.5 and if it is 2 or 3 then it cannot be called as half life. This dilemma can be overcome by understanding that all formula relating to kinetics are only valid when the sample size is very large and in such a large sample size ,a small difference of 0.5 will be insignificant.

The fact that radioactive decay follows the exponential law implies that this phenomenon is statistical in nature. Every nucleus in a sample of a radionuclide has a certain probability of decaying, but there is no way to know in advance which nuclei will actually decay in a particular time span. If the sample is large enough – that is, if many nuclei are present – the actual fraction of it that decays in a certain time span will be very close to the probability for any individual nucleus to decay. To say that, a certain radioisotope has a half –life of 5 hr., then, signifies that every nucleus of this isotope has a 50 percent chance of decaying in every 5 hr. period. This does not mean a 100 percent probability of decaying in 10 hr. A nucleus does not have a memory, and its decay probability per unit time is constant until it actually does decay. A half life of 5 hr. implies a 75% probability of decay in 10 hr., which increases to 87.5% in 15 hr, to 93.75% in 20 hr, and so on, because in every 5 hr. interval the probability of decay is 50 percent.

8.2 Average – Life Time

Average life time is defined as the life time of a single isolated nucleus. Let us imagine a single nucleus which decays in 1 second. Assuming 1 second time interval to be very small the rate of change of nuclei would be 1/1 (because –dN = 1 and dt =1). We can also see that since $\frac{-dN}{dt} = \lambda N$, for a single isolated nucleus N = 1, $\frac{-dN}{dt} = \lambda$. Therefore in this present case λ=1.

Now let us assume the same nucleus decays in 2 seconds, we can see that $\frac{-dN}{dt}$ i.e., λ is equal to ½. You will also notice that in the 1st case the nucleus survived for 1 second and in the second case it survived for 2 second. Therefore the life time of a single isolated nucleus is $\frac{1}{\lambda}$.

$\therefore\ t_{av} = \frac{1}{\lambda}$

8.3 Activity

Activity by definition is the rate of decay of a radioactive element. It is represented as 'A' and is equal to λN. By no means should activity be confused with rate of change of radioactive nuclei represented by $\frac{-dN}{dt}$. This is because $\frac{-dN}{dt}$ talks about the overall change in the number of nuclei in a given instant of time while activity only talks about that change which is decay. For example if you go out to a market with Rs. 50 in your pocket and you spend Rs.20 in 5 minutes then your rate of change of money in the wallet is Rs. 4 / min and in fact the rate of spending the money is also Rs.4 / min. Here you can see both are same. But if while spending Rs. 20 in 5 minutes, somebody keeps Rs.10 in your wallet, then the rate of change of money in your wallet would become Rs. 2.5 /min, while the rate of spending the money is Rs. 4 / min. This implies that as long as the radioactive substance is only decaying the rate of change of nuclei and activity are same and equation (1) in terms activity of a radioactive substance can be written as $A_t = A_o e^{-\lambda t}$. But if the radioactive substance is also being produced this $\frac{dN}{dt}$ = rate of production – activity, (Of course it's a different matter, rate of production may or may not be a constant).

8.4 Specific Activity

This is activity per unit mass of the sample. Let radioactive sample weighing w gms have a decay constant λ. The number of nuclei in the w gms would be $\frac{w}{M} \times Av$, where M = molecular weight of the radioactive substance and Av= Avogadro's number.

$$\therefore \text{ specific activity} = \frac{\left(\lambda \times \frac{w}{M} \times Av\right)}{w} = \frac{\lambda \times Av}{M}$$

It should be remembered that if a radioactive sample is pure and the product does not remain with reactant then specific activity is a constant.

Units of Activity

The unit of radioactivity of a substance is measured as the rate at which it changes into daughter nucleus. It has been derived on the scale of disintegration of Radium.

Let us consider 1g of radium (atomic mass = 226 and $t_{½}$=1600 yrs) undergoes decay, then

Rate of decay of radium = $\lambda \times$ Number of nuclei of Ra in 1 g

$$= \frac{0.693}{1600 \times 365 \times 24 \times 60 \times 60} \times \frac{1 \times 6.023 \times 10^{23}}{226} = 3.7 \times 10^{10} \text{ dps} = 3.7 \times 10^{10} \text{ Becquerrel.}$$

= 1 curie ($\because$ 1 Ci = 3.7×10^{10} dps)

= 3.7×10^4 Rutherford ($\because$ 1 rd = 10^6 dps)

The SI unit of activity is dps or Becquerrel.

8.5 Carbon Dating and Rock Dating

Carbon Dating: The intensity of cosmic ray and hence of ^{14}C in the atmosphere has been remaining constant over thousands of years. In living material the ratio of ^{14}C to ^{12}C remains relatively constant. When the tissue in an animal or plant dies assimilation of radioactive C^{14} ceased to continue. Therefore, in the dead tissue the ratio of ^{14}C to ^{12}C would decrease depending on the age of the tissue.

$$^{14}_{7}N + ^{1}_{0}n \longrightarrow ^{14}_{6}C + ^{1}_{1}p$$
$$^{14}_{6}C \longrightarrow ^{14}_{7}N + ^{0}_{-1}e$$

A sample of dead tissue is burnt to carbondioxide and carbondioxide is analysed for the ratio of ^{14}C to ^{12}C. From this data age of dead tissue (plant or animal) can be determined.

$$\text{Age(t)} = \frac{2.303}{\lambda}\log\frac{N_0}{N}$$
$$\Rightarrow \text{Age} = \frac{2.303 \times t_{1/2}(^{14}C)}{0.693}\log\left(\frac{N_0}{N}\right)$$

N_0 = ratio of $^{14}C/^{12}C$ in living plant
N = ratio of $^{14}c/^{12}C$ in the wood

$$\text{Also} = \frac{2.303 \times t_{1/2}}{0.693}\log\left(\frac{A_0}{A}\right)$$

A_0 = original activity
A = final activity

$$\text{Also } N = \left(\frac{1}{2}\right)^n N_0$$
$$\text{Where } n = \frac{t}{t_{1/2}}$$

Rock Dating: It is based on the kinetic of radioactive decay. It is assumed that no lead was originally present in the sample whole of it came from uranium.

Initial no. of mole (N_0) = [U] + [Pb]

Final no. of mole (N) = [U]

$$\frac{N_0}{N} = \frac{[U]+[Pb]}{[U]} = 1 + \frac{[Pb]}{[U]}$$
$$t = \frac{2.303}{\lambda}\log\left[\frac{N_0}{N}\right]$$
$$\Rightarrow t = \frac{2.303}{\lambda}\log\left[1 + \frac{[Pb]}{[U]}\right]$$

Also $$\left[1 + \frac{[Pb]}{[U]}\right] = (2)^n \quad \left(n = \frac{t}{t_{1/2}}\right)$$

Illustration 5: ***The amount of $^{14}_{6}C$ isotope in a piece of wood is found to be one fifth of that present in a fresh piece of wood. Calculate the age of the piece of wood (half life $^{14}_{6}C = 5577$ year)***

Solution: $$t = \frac{2.303}{\lambda}\log\frac{N_0}{N}$$

$$\Rightarrow t = \frac{2.303 \times t_{1/2}}{0.693} \log \frac{N_0}{N}$$

$$= \frac{2.303 \times 5577}{0.693} \log \frac{N_0}{N_0 / 5}$$

$$= \frac{2.303 \times 5577}{0.693} \times 0.6989 = 12.953 \text{years}$$

Illustration 6: ***The activity of the hair of an Egyptian mummy is 7 disintegration minute^{-1}. Find out age of mummy. Given $t_{1/2}$ of ^{14}C is 5770 year and disintegration rate of fresh sample of ^{14}C is 14 disintegration minute^{-1}.***

Solution: r_0 = 14 dpm

R = 7 dpm

$$\frac{r_0}{r} = 2$$

Rate at any time α no. of atoms at that time

$$\therefore \frac{r_0}{r} = \frac{N_0}{N} = 2$$

$$\therefore t = \frac{2.303}{\lambda} \log \frac{N_0}{N}$$

$$= \frac{2.303 \times 5770}{0.693} \log 2 = 5770 \text{ years}$$

Illustration 7: ***A sample of uranite, a uranium containing mineral, was found on analysis to contain 0.214 gm of ^{206}Pb for every gm of ^{238}U. If all the lead came from the radioactive disintegration of Uranium and assume that all isotopes of uranium other than ^{238}U can be neglected, estimate the date when the mineral was formed in the earth crust. (The half–life of ^{238}U is 4.5×10^9 years).***

Solution:

$$[Pb] = \frac{0.214}{206} = 1.04 \times 10^{-3} \text{mol}$$

$$[U] = \frac{1}{238} = 4.2 \times 10^{-3} \text{mol}$$

$$\frac{[Pb]}{[U]} = 0.247$$

$$\text{As } \left[1 + \frac{Pb}{U}\right] = (2)^{t/t_{1/2}}$$

$$\Rightarrow [1 + 0.247] = 2^{t/t_{1/2}}$$

$$\Rightarrow \frac{t}{t_{1/2}} = 0.3188$$

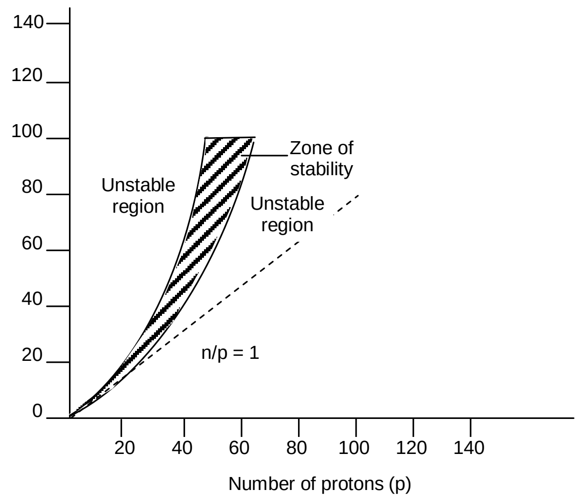
$$\Rightarrow t = 0.3188 \times t_{1/2}$$

$$= 0.3188 \times 4.5 \times 10^9$$

$$1.4344 \times 10^9 \text{ years}$$

8.6 Stability of Nuclei with respect to neutron–proton ratio

If number of neutrons is plotted against the number of protons, the stable nuclei lie within well–defined region called zone of stability.



All the nuclei falling outside this zone are invariably radioactive and unstable in nature Nuclei that fall above the stability zone has an excess of neutrons while those lying below have more protons. These nuclei attain stability making adjustment in n/P ratio.

(A) when (n/p) ratio is higher than that required for stability: Such nuclei have tendency to emit β–rays (transforming a neutron into proton.

$$^{1}_{0}n \longrightarrow {}^{14}_{1}p + {}^{0}_{-1}e(\beta - \text{particle})$$

$$\text{e.g. } {}^{14}_{6}C \longrightarrow {}^{14}_{7}N + {}^{0}_{-1}e$$

$$\frac{n}{p} \quad 1.33 \qquad 1$$

$$^{87}_{36}Kr \longrightarrow {}^{87}_{37}Rb + {}^{e}_{-1}e$$

$$\frac{n}{P} : \frac{51}{36} \qquad \frac{50}{37}$$

(B) when (n/p) ratio is lower than that required for stability: Such nuclei have tendency to increase n/P ratio by adopting any of the following three ways.

i) By emission of an α–particle (Natural radioactivity).

$$^{298}_{92}U \longrightarrow {}^{234}_{90}Th + {}^{4}_{2}He\ (\alpha - \text{particle})$$

$$\left(\frac{n}{P}\right) \quad \left(\frac{146}{92}\right) \qquad \left(\frac{144}{90}\right)$$

$$= 1.50 \qquad = 1.60$$

ii) By emission of positron

$$^{13}_{7}N \longrightarrow ^{13}_{6}C + ^{0}_{+1}e$$

$$\frac{n}{P} \qquad \frac{6}{7} \qquad \qquad \frac{7}{6}$$

iii) By K–electron capture

$$^{194}_{79}Au + ^{0}_{-1} \longrightarrow ^{194}_{78}Pt$$

$$\left(\frac{n}{p}\right) \qquad \frac{115}{79} \qquad \qquad \frac{116}{78}$$

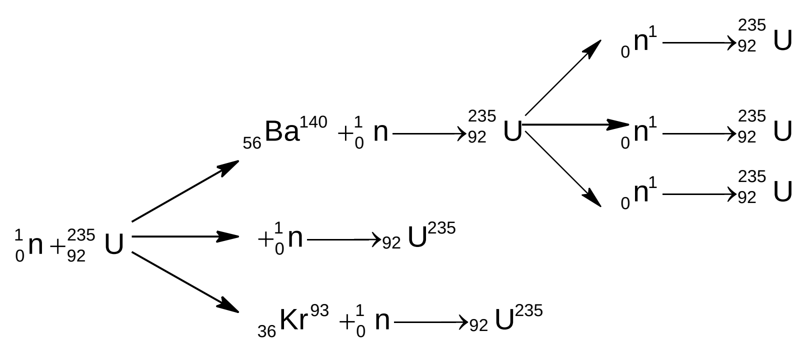
α–emission is usually observed in natural radioactive isotopes while emission of positron or K–electron capture is observed in artificial radioactive isotopes. The unstable nuclei continue to emit α^- or β^- particles until stable nucleus comes into existence

8.7 Nuclear Fission

It is a nuclear reaction in which heavy nucleus splits into lighter nuclei of comparable masses with release of large amount of energy by bombardment with suitable subatomic particles i.e.

$$^{1}_{0}n + ^{235}_{92}U \longrightarrow \begin{cases} {}_{56}Ba^{140} + {}_{36}Kr^{93} + 3\,{}_{0}n^{1} \\ {}_{54}Xe^{144} + {}_{38}Sr^{90} + 2\,{}_{0}n^{1} \\ {}_{55}Cs^{144} + {}_{37}Rb^{90} + 2\,{}_{0}n^{1} \end{cases}$$

If the neutrons from each nuclear fission are absorbed by other ${}_{92}U^{235}$ nuclei, these nuclei split and release even more neutrons. Thus a chain reaction can occur. A nuclear chain reaction is a self sustaining series of nuclear fissions caused by the previous neutrons released from the previous nuclear reactions.



There should be critical size of the fissionable material to maintain fission chain. This in turn requires, minimum critical mass of the fissionable material. It is the small mass of the fissionable material in which chain reaction can be sustained. If mass is larger than this mass (supercritical mass) number of nuclei that split multiply rapidly. An atomic bomb is detonated with small amount of chemical explosive that push together two or more masses of fissionable material to get a supercritical mass.

A nuclear fission reactor is a device that permits a controlled chain nuclear fissions. Control rods made of elements such as boron and cadmium, absorb additional neutrons and can therefore, slow the chain reactions.

8.8 Nuclear Fusion

It is a nuclear reaction in which two lighter nuclei are fused together to form a heavier nuclei. To achieve this, colliding nuclei must posses enough kinetic energy to overcome the initial force of repulsion between the positively charged core. At very high temperature of the order of 10^6 to 10^7 K, the nuclei may have the sufficient energy to overcome the respulsive forces and fuse. Such reactions are therefore also known as thermonuclear reactions.

$$_1^2H + _1^3H \longrightarrow _2^4He + _0^1n + 17.8\text{ MeV}$$
$$_1^2H + _1^2H \longrightarrow _2^4He + 24.9\text{ MeV}$$
$$_1^1H + _1^3H \longrightarrow _2^4He + 20.0\text{ MeV}$$
$$_3^7Li + _1^1H \longrightarrow _2^4He + 17.7\text{ MeV}$$

The energy of fusion process is due to mass defect (converted into binding energy). The high temperature required to initiate such reaction may be attained initially through fission process.

Hydrogen bomb is based on the principle of fusion reactions. Energy release is so enormous that it is about 1000 time that of atomic bomb. In hydrogen bomb a mixture of deuterium oxide (D_2O) and tritium oxide (T_2O) is enclosed in space surrounding an ordinary atomic bomb. The temperature produced by the explosion of the atomic bomb initiates the fusion reaction between $_1^2H$ and $_1^3H$ releasing huge amount of energy.

It is believed that the high temperature of stars including the sum is due to fusion reactions. E. Salt Peter in 1953, proposed a proton–proton chain reaction.

$$_1^1H + _1^1H \longrightarrow _1^2H + _{+1}^0e + \gamma$$
$$_1^2H + _1^1H \longrightarrow _2^3He + \gamma$$
$$_2^3He + _1^1H \longrightarrow _2^4He + _{+1}^0e + \gamma$$

$$4_1^1H \longrightarrow _2^4He + 2_{+1}^0e + 24.7\text{MeV}$$

9. Solutions to Exercises

Exercise 1:

a)

	$A \longrightarrow$	Product
Initially	C_o	0
At time t	$C_o(1-\alpha)$	αC_o

Concentration of A, at time t (C) = $C_o(1-\alpha)$

For 1st order reaction $K = \frac{1}{t}\ln\frac{C_o}{C}$

$$Kt = \ln\frac{C_o}{C} = \ln\frac{C_o}{C_o(1-\alpha)}$$

$$Kt = \ln\frac{1}{(1-\alpha)} \text{ or } \frac{1}{1-\alpha} = e^{Kt}$$

$1-\alpha = e^{-Kt}$

or $\alpha = 1 - e^{-Kt}$

b) The rates of formation of products must be of the same order of magnitude. If they were very different, the rate of the overall reaction would reflect only the faster mechanism.

Exercise 2:

Rate I $= K'_1 = [A][B] = 1\times10^{-5}\times0.1 = 1.0\times10^{-7}$ M sec^{-1}

Rate II $= K_1[A] = (1\times10^{-4}\ s^{-1})(0.1) = 1\times10^{-5}$ M sec^{-1}

$K'_1[A][B] = K_1[A]$

$$[B] = \frac{K_1}{K'_1} = \frac{1\times10^{-4}s^{-1}}{1\times10^{-5}M^{-1}s^{-1}} = 10\ M$$

Exercise 3:

a) At steady state, rate of destruction of O_3 must be equal to the rate of its generation i.e. 7.2×10^{-15} mol/L per hour.

From second order rate law

$$K = \frac{1}{[O_3]^2}\frac{\Delta[O_3]}{\Delta t} = \frac{1}{2.0\times10^{-8}}\frac{7.2\times10^{-15}}{3.6\times10^{3}} = 5\times10^{-3}\ L\ mol^{-1}\ s^{-1}$$

b) $$\text{Slope} = \frac{(2.9-1.1)}{(1.3-1.5)\times10^{-3}K^{-1}} = -9\times10^{3}K$$

$E_a = -2.303\ R\ (\text{slope}) = -2.303\ (1.99)\ (-9\times10^3)$

$= 4\times10^4$ cal/mol = 40 kcal/mol

Exercise 4:

Taking clue from the previous illustrations

$P_1 = P_3$ and $x = P_2$

$\therefore P_1 - x = P_3 - P_2$

$$\therefore\ \mathbf{k = \frac{1}{t}\ln\frac{[A]_0}{[A]_t} = \frac{l}{t}\ln\frac{P_3}{(P_3-P_2)}}$$

Exercise 5:

	$A \longrightarrow$	B +	C
At t = 0	V_1	0	0
At t = t	$V_1 - x$	$2x$	$3x$

$\therefore V_1 + 4x = V_2$

$\Rightarrow x = \dfrac{V_2 - V_1}{4} \Rightarrow V_1 – x = V_1 – \dfrac{V_2}{4} + \dfrac{V_1}{4} = \dfrac{5V_1 - V_2}{4}$

$\therefore$ **k =** $\dfrac{1}{t}\ln\dfrac{[A]_0}{[A]_t} = \dfrac{1}{t}\ln\dfrac{4V_1}{(5V_1 - V_2)}$

Exercise 6:

$V_1 = V_D$

$V_2 = 2x + V_D$

$V_3 = 2V_A + V_D$

$\therefore \dfrac{V_3 - V_1}{2} = V_A$

$\dfrac{V_3 - V_2}{2} = V_A - x$

$\therefore k = \dfrac{1}{t}\ln\dfrac{[A]_0}{[A]_t} = \dfrac{1}{t}\ln\dfrac{(V_3 - V_1)}{(V_3 - V_2)}$

Exercise 7:

$V_1 = V_A + V_D$

$V_2 = (V_A – x) + 2x + 3x + V_D$

$V_3 = 2V_A + 3V_A + V_D$

We can see that $\dfrac{V_3 - V_1}{4} = V_A$

and $\dfrac{V_3 - V_2}{4} = V_A - x$

$\therefore$ **k** $= \dfrac{1}{t}\ln\dfrac{[A]_0}{[A]_t} = \dfrac{1}{t}\ln\dfrac{(V_3 - V_1)}{(V_3 - V_2)}$

***Exercise 8*:**

a) Let the rotation of Glucose be r_2^0 and that of fructose be $-r_3^0$ (since it is leavo–rotatory) per mole.

$\therefore$	S $\longrightarrow$	G +	F
At t = 0	a	0	0
At t = t	a– x	x	x
At t = ∞	0	a	a

$\therefore r_t = x\left(r_2^o - r_3^o\right)$

$r_\infty = a\left(r_2^o - r_3^o\right)$

$a = \dfrac{r_\infty}{r_2^o - r_3^o}$, $x = \dfrac{r_t}{r_2^o - r_3^o}$

$\therefore a – x = \dfrac{r_\infty - r_t}{r_2^o - r_3^o}$

$\therefore$ **k** $= \dfrac{1}{t}\ln\dfrac{a}{a - x} - = \dfrac{1}{t}\ln\dfrac{r_\infty}{(r_\infty - r_t)}$

b)

	S $\longrightarrow$	G +	F
At t = 0	a	0	0
At t = t	a– x	x	x
At t = ∞	0	a	a

$\therefore$ $r_0 = a\ r_1^o$ …(1)

$r_t = (a - x)\ r_1^o + x$ …(2)

$r_\infty = a\left(r_2^o - r_3^o\right)$ …(3)

Simplifying equation (2)

$r_t = a r_1^o - x\left(r_1^o - r_2^o + r_3^o\right)$

Since $a r_1^o$, is r_0

$\therefore x = \dfrac{r_0 - r_t}{\left(r_1^o - r_2^o + r_3^o\right)}$

Now, we can write 'a' as $\dfrac{r_o}{r_1^o}$ or $\dfrac{r_\infty}{r_2^o - r_3^o}$. But the problem in writing either of them is that the constants of a and x are different. This implies that the constants of a–x would also be different from that of 'a'.

So we write, $r_o - r_\infty = a\left(r_1^o - r_2^o + r_3^o\right)$

$\therefore a = \dfrac{r_0 - r_\infty}{\left(r_1^o - r_2^o + r_3^o\right)} \Rightarrow a - x = \dfrac{r_t - r_\infty}{\left(r_1^o - r_2^o + r_3^o\right)}$

$\therefore \mathbf{k} = \dfrac{\mathbf{1}}{\mathbf{t}}\mathbf{ln}\dfrac{\mathbf{a}}{\mathbf{a} - x} = \dfrac{\mathbf{1}}{\mathbf{t}}\mathbf{ln}\dfrac{\left(\mathbf{r_0} - \mathbf{r_\infty}\right)}{\left(\mathbf{r_t} - \mathbf{r_\infty}\right)}$

10. Solved Problems

10.1 Subjective

Problem 1: ***Two reactions of the same order have same exponential factors but their activation energies differ by 24.9 kg/mol. Calculate the ratio of between the rate constants of these reactions at 27°C (R = 8.31 JK^{-1} mol^{-1}).***

Solution:

$$\log_{10} K = \log_{10} A - \frac{E}{2.303RT}$$

$$\log_{10} K_1 = \log_{10} A - \frac{E_1}{2.303RT}$$

$$\log_{10} K_2 = \log_{10} A - \frac{E_2}{2.303RT}$$

$$\log_{10}\left(\frac{K_2}{K_1}\right) = \frac{(E_1 - E_2)}{2.303RT} \Rightarrow \log_{10}\frac{K_2}{K_1} = \frac{24.9 \times 1000}{2.303 \times 8.314 \times 300}$$

$$\therefore \frac{K_2}{K_1} = 2.199 \times 10^4$$

Problem 2: ***In Arrhenius equation for a certain reaction, the values of A, and E_a (activation energy) are 4×10^{13} sec^{-1} and 98.6 kJ mol^{-1} respectively. If the reaction is of first order at what temperature will its its half–life period is 10 minutes?***

Solution:

$$K = Ae^{-Ea/RT}$$

$$\log_e K = \log_e A - \frac{Ea}{RT} \Rightarrow \log_{10} K = A - \frac{Ea}{2.303RT}$$

For first order reaction $t_{1/2} = \frac{0.693}{K}$

$$K = \frac{0.693}{600}\text{sec}^{-1} \ (t_{1/2} = 10 \text{ minute} = 600 \text{ sec})$$

$$= 1.1 \times 10^{-3} \text{ sec}^{-1}$$

$$\therefore \log_{10}(1.1 \times 10^{-3}) = \log_{10}(4 \times 10^{13}) - \frac{98.6 \times 1000}{2.303 \times 8.314 \times T}$$

$$\therefore T = 310.95K$$

Problem 3: ***The rate constant for an isomerisation reaction $A \rightarrow B$ is 4.5×10^{-3} min^{-1}. If the initial concentration of A is 1 M calculate the rate of reaction after 1 hour.***

Solution:

$A \rightarrow B$, K = 4.5×10^{-3} min^{-1}, a = 1M

$$K = \frac{2.303}{t}\log_{10}\frac{a}{(a-x)}$$

$$\Rightarrow 4.5 \times 10^{-3} = \frac{2.303}{60}\log\frac{1}{(a-x)} \Rightarrow (a-x) = 0.7634$$

Thus rate after 60 minute = K (a – x)

= $4.5 \times 10^{-3} \times 0.7634$ = 3.4354×10^{-3}

Problem 4: ***For the reaction A + B $\longrightarrow$ C the following data were obtained. In the first experiment when the initial concentrations of both A and B are 0.1 M the observed initial rate of formation of C is 1 $\times$ 10^{-4} mol lt^{-1} $minute^{-1}$. In second experiment when the initial concentrations of (A) and(B) are 0.1 M and 0.3 M the initial rate is 9.0 $\times$ 10^{-4} mol lt^{-1} $minute^{-1}$. In the third experiment when the initial concentrations of both A and B are 0.3 M the initial rate is 2.7 $\times$ 10^{-3} mol $litre^{-1}$ $minute^{-1}$.***

a) Write rate law for this reaction.

b) Calculate the value of specific rate constant for this reaction.

Solution: Let Rate = $K[A]^x[B]^y$

$r_1 = 1\times10^{-4} = K(0.1)^x(0.1)^y \qquad \ldots(1)$

$r_2 = 9\times10^{-4} = K(0.1)^x(0.3)^y \qquad \ldots(2)$

$r_3 = 2.7\times10^{-3} = K(0.3)^x(0.3)^y \qquad \ldots(3)$

By equations (1) and (2) $\frac{r_1}{r_2} = \frac{1\times10^{-4}}{9\times10^{-4}} = \left(\frac{1}{3}\right)^y \therefore y = 1$

By equations (2) and (3) $\frac{r_2}{r_3} = \frac{9\times10^{-4}}{27\times10^{-4}} = \left(\frac{1}{3}\right)^x \therefore x = 1$

$\therefore \text{Rate} = K[A]^1[B]^2$

Also $1\times10^{-4} = K(0.1)^1(0.1)^2$

$\therefore K = 10^{-1} = 0.1 Lt^2 mol^{-1} min^{-1}$

Problem 5: ***Rate constant of a first order reaction A $\to$ B is 0.0693 min^{-1}. Calculate rate (i) at start (ii) after 30 minutes. Initial concentration of A is 1.0 M (Don't use log table).***

Solution: Rate constant of the first order reaction (K_1) = 0.0693 min^{-1}.

$$t_{1/2} = \frac{0.693}{K_1} = \frac{0.693}{0.0693} = 10\,\text{min}$$

$$C = C_0\left(\frac{1}{2}\right)^n \left(n = \frac{\text{total time}}{t_{1/2}} = \frac{30}{10} = 3\right)$$

$$C_0 = 1M, \therefore C = 1\times\left(\frac{1}{2}\right)^3 = \frac{1}{8}M$$

Rate of reaction at the start of reaction = K_1C_0

$= 0.0693\times1 = 6.93\times10^{-2} M/min$

$\therefore$ Rate of reaction after 30 minute = K, C = $8.66\times10^{-3} M/min$

Problem 6: ***A hydrogenation reaction is carried out at 500K. If the same reaction is carried in presence of a catalyst at the same rate the temperature required is 400K. Calculate the activation energy of the reaction if the catalyst lowers the activation barrier by 20 kJ/mol.***

Solution: $K = Ae^{-Ea/RT}$ (Arrhenius equation)

Let rate constants at temperatures 500K and 400K are K_{500} are K_{400} respectively.

$K_{500} = Ae^{-E_{500}/R\times 500}$ …(1)

$K_{400} = Ae^{-E_{400}/R\times 400}$ …(2)

Given $K_{500} = K_{400}$ (same rate in presence and absence of catalyst)

From equation (1) and (2)

$$\frac{E_{500}}{R\times 400} = \frac{E_{400}}{R\times 400} \Rightarrow \frac{E_{500}}{5} = \frac{E_{400}}{4}$$

$$E_{500} = \frac{E_{400}}{4}\times 5$$

Given $E_{500} = E_{400} + 20$ …(3)

$\therefore E_{400} + 20 = E_{400}\times 1.25$

$$\therefore E_{400} = \frac{200}{0.25} = 80\text{kJ/mol}$$

$E_{500} = 80 + 20 = 100\text{kJ/mol}$

Problem 7: $^{238}_{92}U$ ***by successive radioactive decay changes to*** $^{206}_{82}Pb$. ***A sample of uranium ore was analysed and found to contain 1.0 gm of*** ^{238}U ***and 0.1 gm of*** ^{206}Pb. ***Assuming that all the*** ^{206}Pb ***has accumulated due to decay of*** ^{238}U***, find the age of the ore (half–life of*** $^{238}U = 4.5\times 10^{9}$ ***years).***

Solution: No. of moles of $^{238}U = \frac{1}{238}$

No. of moles of $^{206}Pb = \frac{0.1}{206}$

$$t = \frac{2.033}{\lambda}\log\left[1+\frac{^{206}Pb}{^{238}U}\right]$$

$$= \frac{2.303}{0.693}\times 4.5\times 10^{9}\log\left[1+\frac{\frac{0.1}{0.206}}{\frac{1}{208}}\right]$$

$$= \frac{2.303}{0.693}\times 4.5\times 10^{9}\log(1.1155)$$

$$= 7.098\times 10^{8}\text{ years}$$

Problem 8: ***For a first order reaction when log k was plotted against 1/T, a straight line with a slope of -6000 was obtained. Calculate the activation energy of the reaction***

Solution: Slope = $\frac{-Ea}{2.303R}$ = –6000

$Ea = 6000\times 2.303\times 8.314 = 1.148\times 10^{5}J$

$$= \frac{1.148\times 10^{5}J}{4.18J/cal} = 27483.935\text{Cal} = 27.48\text{ Kcal}$$

Problem 9: ***For the first order decomposition reaction***

$(CH_3)_3COOC(CH_3)_3 \longrightarrow 2CH_3COCH_3 + C_2H_6$

in the gaseous phase, the pressure of the system at t = 0 and t = 15 min were found to be 169.3 torr and 256.0 torr, respectively. Calculate (a) the rate constant of the reaction, (b) half-life period and (c) the pressure of the system at 9 min.

Solution: $(CH_3)_2COOC(CH_3)_3 \longrightarrow 2CH_3COCH_3 + C_2H_6$

P_0	0	0	t = 0
$P_0 - P$	2P	P	t = t

Total pressure at time t, $P_t = P_0 + 2P$

So, $256.0 = 169.3 + 2P$, $\therefore$ P = 43.35 torr, and

$P_0 - P = 169.3 - 43.35 = 125.95$ torr

a) Now using the integrated rate expression for the first order reaction
We get,
$$k = \frac{2303}{15}\log\frac{169.3}{125.95} = 0.0197\ \text{min}^{-1}$$

b) The half-life period is
$$t_{1/2} = \frac{0.693}{k} = \frac{0.693}{0.0197} = 31.18\ \text{min}$$

c) Substituting the known parameters is the integrated rate expression,
we get,
$$0.0197 = \frac{2.303}{9}\log\frac{169.3}{169.3 - P}$$
Upon solving: P = 27.50 torr
$\therefore P_t = 169.23 + 2 \times 27.50 = 224.3$ torr

Problem 10: ***From the following data show that the decomposition of hydrogen peroxide in aqueous solution is a first - order reaction. What is the value of the rate constant?***

Time in minutes	***0***	***10***	***20***	***30***	***40***
Volume V in ml	***25.0***	***20.0***	***15.7***	***12.5***	***9.6***

where V is the number of ml of potassium permanganate required to decompose a definite volume of hydrogen peroxide solution.

Solution: The equation for a first order reaction is

The volume of $KMnO_4$ used, evidently corresponds to the undecomposed hydrogen peroxide.

Hence the volume of $KMnO_4$ used, at zero time corresponds to the initial concentration a and the volume used after time t, corresponds to (a - x) at that time. Inserting these values in the above equation, we get

when t = 10 min. $k_1 = \frac{2.303}{10}\log\frac{25}{20.0} = 0.022318\ \text{min}^{-1} = 0.000372\ \text{s}^{-1}$

when t = 20 min. $k_1 = \frac{2.303}{20}\log\frac{25}{15.7} = 0.023265\ \text{min}^{-1} = 0.0003871\ \text{s}^{-1}$

when t = 30 min. $k_1 = \frac{2.303}{30}\log\frac{25}{12.5} = 0.02311\,\text{min}^{-1} = 0.000385\ s^{-1}$

when t = 40 min. $k_1 = \frac{2.303}{40}\log\frac{25}{9.6} = 0.023932\,\text{min}^{-1} = 0.0003983\ s^{-1}$

The constancy of k, shows that the decomposition of H_2O_2 in aqueous solution is a **first order** reaction.

The average value of the rate constant is **0.0003879 s^{-1}**.

Problem 11: ***5 ml of ethylacetate was added to a flask containing 100 ml of 0.1 N HCl placed in a thermostat maintained at 30°C. 5 ml of the reaction mixture was withdrawn at different intervals of time and after chilling, titrated against a standard alkali. The following data were obtained :***

Time (minutes)	***0***	***75***	***119***	***183***	***∞***
Volume of alkali used in ml	***9.62***	***12.10***	***13.10***	***14.75***	***21.05***

Show that hydrolysis of ethyl acetate is a first order reaction.

Solution: The hydrolysis of ethyl acetate will be a first order reaction if the above data confirm to the equation.

$$k_1 = \frac{2.303}{t}\log\frac{V_\infty - V_0}{V_\infty - V_t}$$

Where V_0, V_t and V_∞ represent the volumes of alkali used at the commencement of the reaction, after time t and at the end of the reaction respectively, Hence

$V_\infty - V_0 = 21.05 - 9.62 = 11.43$

Time **V_∞ - V_t** **k_1**

75 min 21.05 - 12.10 = 8.95 $\frac{2.303}{75}\log\frac{11.43}{8.95} = 0.003259\ \text{min}^{-1}$

119 min 21.05 - 13.10 = 7.95 $\frac{2.303}{119}\log\frac{11.43}{7.95} = 0.003051\,\text{min}^{-1}$

183 min 21.05 - 14.75 = 6.30

$\frac{2.303}{183}\log\frac{11.43}{6.30} = 0.003254\ \text{min}^{-1}$

A constant value of k shows that hydrolysis of ethyl acetate is a **first order** reaction

Problem 12: ***The optical rotations of sucrose in 0.5 N HCl at 35°C at various time intervals are given below. Show that the reaction is of first order :***

Time (minutes)	***0***	***10***	***20***	***30***	***40***	***∞***
Rotation (degrees)	***+32.4***	***+28.8***	***+25.5***	***+22.4***	***+19.6***	***-11.1***

Solution: The inversion of sucrose will be first order reaction if the above data confirm to the equation , $k_1 = \frac{2.303}{t}\log\frac{r_0 - r_\infty}{r_t - r_\infty}$

Where r_0, r_t and r_∞ represent optical rotations initially, at the commencement of the reaction after time t and at the completion of the reaction respectively

In the case $a_0 = r_0 - r_\infty = +32.4 - (-11.1) = +43.5$

The value of k at different times is calculated as follows :

Time	**r_t**	**$r_t - r_\infty$**	**k**
10 min	+28.8	39.9	$\frac{2.303}{10}\log\frac{43.5}{39.9} = 0.008625\ min^{-1}$
20 min	+25.5	36.6	$\frac{2.303}{20}\log\frac{43.5}{36.6} = 0.008625\ min^{-1}$
30 min	+22.4	33.5	$\frac{2.303}{30}\log\frac{43.5}{33.5} = 0.008694\ min^{-1}$
40 min	+19.6	30.7	$\frac{2.303}{40}\log\frac{43.5}{30.7} = 0.008717\ min^{-1}$

The constancy of k_1 indicates that the inversion of sucrose is a **first order** reaction.

Problem 13: ***The reaction given below, involving the gases is observed to be first order with rate constant 7.48 $\times$ 10^{-3} sec^{-1}.Calculate the time required for the total pressure in a system containing A at an initial pressure of 0.1 atm to rise to 0.145 atm and also find the total pressure after 100 sec.***

$2A(g) \rightarrow 4B(g) + C(g)$

Solution:

	2A(g) $\longrightarrow$	4B(g)	+	C(g)
initial	P_o	0		0
at time t	$P_o - P'$	$2P'$		$P'/2$

$P_{total} = P_o - P' + 2P' + P'/2 = P_o + \frac{3P'}{2}$

$P' = \frac{2}{3}(0.145 - 0.1) = 0.03$ atm

$k = \frac{2.303}{t}\log\frac{P_o}{P_o - P'}$

$t = \frac{2.303}{7.48\times10^{-3}}\log\left(\frac{0.1}{0.07}\right)$

t = 47.7 sec

Also, $k = \frac{2.303}{t}\log\left(\frac{0.1}{P_o - P'}\right)$

$7.48 \times 10^{-3} = \frac{2.303}{100}\log\left(\frac{0.1}{0.1 - P'}\right)$

$0.1 - P' = 0.047$

$P' = 0.053$

$P_{total} = 0.1 + \frac{3}{2}(0.053)$

$\approx$ **0.180 atm.**

Problem 14: ***For a reaction, the energy of activation is zero. What is the value of rate constant at 300 K if k = 1.6 $\times$ 10^6 s^–1 at 280 K? (R = 8.31 J K^–1 mol^–1)***

Solution: We know,

$$\log \frac{K_{300}}{K_{280}} = \frac{E_a}{2.303R}\left[\frac{T_2 - T_1}{T_2T_1}\right]$$

E_a = energy of activation

Given: $E_a = 0$

$$\therefore \frac{E_a}{2.303R}\left[\frac{T_2 - T_1}{T_2T_1}\right] = 0$$

or $\log \frac{K_{300}}{K_{280}} = 0$ or $\frac{K_{300}}{K_{280}} = 1$

Hence $K_{300} = K_{280}$

Or K_{300} = $\mathbf{1.6 \times 10^6\ s^{-1}}$

Problem 15: ***The specific reaction rate of a reaction is 1 $\times$ 10^–3 min^–1 at 27°C and 2 $\times$ 10^–3 min^–1 at 37°C. Calculate the energy of activation of this reaction and its specific reaction rate at 57°C.***

Solution: We know that

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303R}\left[\frac{T_2 - T_1}{T_1T_2}\right] \text{ or } E_a = \frac{2.303RT_1T_2}{T_2 - T_1}\log\frac{k_2}{k_1}$$

Substituting different values in the equation, we get

$$E_a = \frac{(2.303)(2.0\ \text{cal mol}^{-1}\text{K}^{-1})(300\ \text{K})(310\text{K})}{(310\text{K} - 300\text{K})}$$

$$\log \frac{2\times10^{-3}\ \text{min}^{-1}}{1\times10^{-3}\ \text{min}^{-1}} = (42835.8\ \text{cal mol}^{-1})(\log 2)$$

$= (12893.6\ \text{cal mol}^{-1}) = 12900\ \text{cal mol}^{-1}$

To calculate specific reaction rate at 57°C, we have

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303R}\left[\frac{T_2 - T_1}{T_1T_2}\right] \text{ or } \log k_2 = \log k_1 + \frac{E_a}{2.303R}\left[\frac{T_2 - T_1}{T_1T_2}\right]$$

or $\log k_2 = \log 1 \times 10^{-3}\ \text{min}^{-1}$

$$+ \frac{(12900\ \text{cal mol}^{-1})}{(2.303)(2\ \text{cal mol}^{-1}\text{K}^{-1})}\left[\frac{350\text{K} - 300\text{K}}{330\text{K} \times 300\text{K}}\right] = 3 + 0.4243 = 3.4243$$

or $\mathbf{k_2 = 2.656 \times 10^{-3}\ min^{-1}}$

10.2 Objective

Problem 1: ***The rate constant for the reaction***

$2N_2O_5 \longrightarrow 4NO_2 + O_2$ ***is*** $3.0 \times 10^{-5}\ sec^{-1}$

If the rate is 2.4 $\times 10^{-5}$ ***mol Lt^{-1} sec^{-1} the concentration of N_2O_5 (in mol litre^{-1}) is***

(A) 1.4 ***(B) 1.2***
(C) 0.04 ***(D) 0.8***

Solution: Rate = K[N_2O_5] Hence $2.4 \times 10^{-5} = 3.0 \times 10^{-5}\ [N_2O_5]$

$\Rightarrow [N_2O_5] = 0.8$ mol L^{-1}

$\therefore$ **(D)**

Problem 2: ***$SO_2Cl_2 \rightleftharpoons SO_2 + Cl_2$ is the first order gas reaction with K = 2.2 $\times 10^{-5}$ sec^{-1} at 320°C. The percentage of SO_2Cl_2 decomposed on heating for 90 minutes is***

(A) 1.118 ***(B) 0.1118***
(C) 18.11 ***(D) 11.30***

Solution:

$$K = \frac{2.303}{t}\log\frac{a}{(a-x)} \Rightarrow \log\frac{a}{(a-x)} = \frac{K.t}{2.303}$$

$$\Rightarrow \log\frac{a}{a-x} = \frac{2.2\times10^{-5}\times60\times60}{2.303} = 0.0516$$

Hence $\dfrac{a}{(a-x)} = 1.127 \Rightarrow \dfrac{a-x}{a} = 0.887$

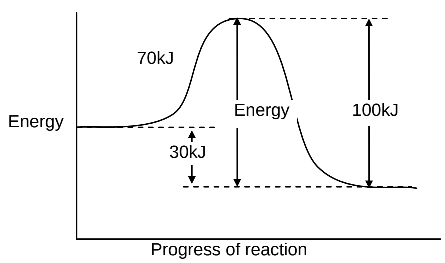
$\Rightarrow 1-\dfrac{x}{a} = 0.887 \Rightarrow \dfrac{x}{a} = 0.113$ = 11.3%

$\therefore$ **(D)**

Problem 3: ***If a reaction A + B $\rightarrow$ C is exothermic to the extent of 30 kJ/mol and the forward reaction has an activation energy 70 kJ/mol, the activation energy for the reverse reaction is***

(A) 30 kJ/mol ***(B) 40kJ/mol***
(C) 70 kJ/mol ***(D) 100 kJ/mol***

Solution:



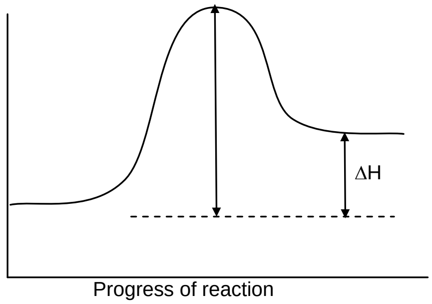
Activation energy for backward reaction = 100 kJ

$\therefore$ **(D)**

Problem 4: ***for an endothermic reaction, where ΔH represents the enthalpy of reaction in kJ mol, the minimum value for the energy of activation will be***

(A) Less than ΔH ***(B) Zero***
(C) More than ΔH ***(D) Equal to ΔH***

Solution:



$E_a > \Delta H$

$\therefore$ **(C)**

Problem 5: ***The rate constant, the activation energy and the Arrhenius parameter of a chemical reaction at 25°C are 3.0×10^{-4} s^{-1}, 104.4 kJ mol^{-1} and 6.0×10^{14} s^{-1} respectively the value of the rate constant as $T \to \infty$ is***

(A) ***2.0×10^{18} s^{-1}*** ***(B)*** ***6.0×10^{14} s^{-1}***
(C) ***∞*** ***(D)*** ***3.6×10^{30} s^{-1}***

Solution: $K = Ae^{-Ea/RT}$
When $T \to \infty$
$K \to A$
$A = 6 \times 10^{14}$ s^{-1}

Problem 6: ***The inversion of cane sugar proceeds with half–life of 500 minute at pH 5 for any concentration of sugar. However if pH = 6, the half–life changes to 50 minute. The rate law expression for the sugar inversion can be written as***

(A) $r = K[sugar]^2[H]^6$ ***(B)*** $r = K[sugar]^1[H^+]^0$
(C) $r = K[sugar]^1[H^+]^1$ ***(D)*** $r = K[sugar]^0[H^+]^1$

Solution: At pH = 5, the half–life is 500 minute for all concentrations of sugar that is $t_{1/2} \propto [sugar]^0$. The reaction is 1st order with respect to sugar.
Let rate = $K[sugar]^1 [H^+]^x$
For $[H^+]$ $t_{1/2} \propto [H^+]^{1-x}$
$\Rightarrow 500 \propto (10^{-5})^{(1-x)}$ $50 \propto (10^{-6})^{1-x}$
$\therefore 10 = 10^{(1-x)} \Rightarrow (1-x) = 1 \therefore x = 0$
Therefore, rate = $K[sugar]^1[H^+]^0$
$\therefore$ **(B)**

Problem 7: ***Two substances A and B are present such that $[A_0] = 4[B_0]$ and half–life of A is 5 minute and that of B is 15 minute. If they start decaying at the same time following first order kinetics how much time later will the concentration of both of them would be same.***

(A) ***15 minute*** ***(B)*** ***10 minute***
(C) ***5 minute*** ***(D)*** ***12 minute***

Solution: Amount of A left in n_1 haves = $\left(\frac{1}{2}\right)^{n_1}[A_0]$

Amount of B left n_2 halves = $\left(\frac{1}{2}\right)^{n_2}[B_0]$

At the end, $\frac{[A^0]}{2^{n_1}} = \frac{[B_0]}{2^{n_2}}$

$\Rightarrow \frac{4}{2^{n_1}} = \frac{1}{2^{n_2}} \left[[A_0] = 4[B_0]\right]$

$\therefore \frac{2^{n_1}}{2^{n_n}} = 4 \Rightarrow 2^{n_1 - n_2} = (2)^2 \therefore n_1 - n_2 = 2$

$\therefore n_2 = (n_1 - 2)$ …(1)

Also $t = n_1 \times t_{1/2(B)}$ $t = n_2 \times t_{1/2(B)}$

(Let concentration both become equal after time t)

$\therefore \frac{n_1 \times t_{1/2(A)}}{n_2 \times t_{1/2(B)}} = 1 \Rightarrow \frac{n_1 \times 5}{n_2 \times 15} = 1 \Rightarrow \frac{n_1}{n_2} = 3$ …(2)

For equation (1) and (2)

$n_1 = 3$, $n_2 = 1$

$t = 3 \times 5 = 15$ minute

$\therefore$ **(A)**

Problem 8: ***The rate of reaction is doubled for every 10° rise in temperature. The increase in reaction rate as result of temperature rise from 10° to 100° is***

(A) 112 ***(B) 512***

(C) 400 ***(D) 614***

Solution: Increase in steps of 10° has been made 9 times. Hence rate of reaction should increase 2^9 times i.e. 512 times.

$\therefore$ **(B)**

Problem 9: ***In uranium mineral, the atomic ratio $\frac{N^{U-238}}{N^{Pb-206}}$ is nearly equal to one. The age (in years) of the mineral is nearly (given that half–life of U^{238} is 4.5×10^9 years).***

(A) 3.0×10^8 ***(B) 4.5×10^8***

(C) 3.0×10^9 ***(D) 4.5×10^9***

Solution: Equal no. of atoms of U^{238} and Pb^{206} means half of U has distintegrated

$\Rightarrow$ time taken = $t_{1/2}$

$\therefore$ **(D)**

Problem 10: ***In the nuclear reaction $^{235}_{92}U \longrightarrow ^{207}_{82}Pb$, the number of α and β particles lost would be***

(A) 8, 4 ***(B) 6, 2***

(C) 7, 4 ***(D) 4, 3***

Solution: $^{235}_{92}U \longrightarrow ^{207}_{82}Pb + x\,^4_2\alpha + y\,^0_{-1}\beta$

$235 = 207 + 4x \Rightarrow x = 7$

$92 = 82 + 2x - y$ or $y = 4$

$\therefore$ **(C)**

Problem 11: ***Fill in the blank***

$^{235}_{92}U + ^{1}_{0}n \longrightarrow ? + ^{92}_{36}Kr + 3^{1}_{0}n$

(A) $^{141}_{56}Ba$ ***(B)*** $^{139}_{56}Ba$

(C) $^{139}_{54}Ba$ ***(D)*** $^{141}_{54}B$

Solution: 92 + 0 = Z + 36 + 0 $\Rightarrow$ Z = 56

235 + 1 $\longrightarrow$ A + 92 + 34

$\therefore$ A = 144

Missing nucleide is $^{141}_{56}B$

$\therefore$ **(A)**

Problem 12: ***For the reaction A + B $\longrightarrow$ C, it is found that doubling the concentration of A, increases the rate 4 times and doubling the concentration of B doubles the reaction rate. What is the overall order of the reaction?***

(A) $\frac{3}{2}$ ***(B) 4***

(C) 1 ***(D) 3***

Solution: Order with respect to A =– 2, order with respect to B = 1 overall order = 3

$\therefore$ **(D)**

Problem 13: ***Consider a gaseous reaction, the rate of which is given by K[A] [B], the volume of the reaction vessel containing these gases is suddenly reduced to 1/4th of the initial volume. The rate of reaction relative to the original rate would be***

(A) 16/1 ***(B) 1/16***

(C) 8/1 (D) ***1/8***

Solution: By reducing volume to $\frac{1}{4}$th the concentration will become 4 times hence rate 16 times.

$\therefore$ **(A)**

Problem 14: ***The rate of a reaction increases 4-fold when concentration of reactant is increased 16 times. If the rate of reaction is 4 $\times$ 10^{-6} mole L^{-1} S^{-1} mole L^{-1} when concentration of the reactant is 4 $\times$ 10^{-4}, the rate constant of the reaction will be***

(A) ***2×10^{-4} mole$^{1/2}$ L$^{-1/2}$ S^{-1}*** ***(B)*** ***1×10^{-2} S^{-1}***

(C) ***2×10^{-4} mole$^{-1/2}$, L$^{1/2}$ S^{-1}*** ***(D)*** ***25 mole^{-1} L min^{-1}***

Solution: Rate $\propto \sqrt{\text{concen}}$, Rate = $k\sqrt{\text{concn}}$

$$k = \frac{\text{Rate}}{(\text{concen})^{1/2}} = \frac{4\times10^{-6}}{\left(4\times10^{-4}\right)^{1/2}} = \frac{4\times10^{-6}}{2\times10^{-2}} = 2 \times 10^{-4}\ \text{mole}^{1/2}\ \text{L}^{-1/2}\ \text{S}^{-1}$$

$\therefore$ **(A)**

Problem 15: ***A catalyst lowers the activation energy of a reaction from 20 kJ mole^{-1} to 10 kJ mole^{-1}. The temperature at which the uncatalysed reaction will have the same rate as that of the catalysed at 27°C is***

(A) –123°C ***(B) 327°C***
(C) 327°C ***(D) + 23°C***

Solution:

$$\frac{E'a}{T_1} = \frac{E_a}{T_2} = \frac{10}{300} = \frac{20}{T_2}$$

$\therefore T_2 = 600\text{ K} = 327^\circ\text{ C}$

$\therefore$ **(B)**

11. Assignments

11.1 Subjective

LEVEL – I

1. The first order reaction has $k = 1.5 \times 10^6$ per second at 200°C. If the reaction is allowed to run for 10 hours, what percentage of the initial concentration would have changed in the product? What is the half-life period of this reaction?

2. For the reaction :
$C_2H_5I + OH^- \rightarrow C_2H_5OH + I^-$
the rate constant was found to have a value of 5.03×10^{-2} mol^{-1} dm^3 s^{-1} at 289 K and 6.71 mol^{-1} dm^3 s^{-1} at 333 K. What is the rate constant at 305 K.

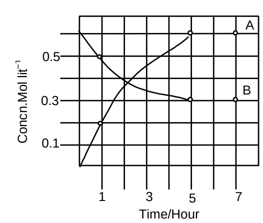
3. At 380°C, the half-life period for the first order decomposition of H_2O_2 is 360 min. the energy of activation of the reaction is 200 kJ mol^{-1}, Calculate the time required for 75% decomposition at 450°C.

4. Fall out from nuclear explosion contains ^{131}I and ^{90}Sr. Calculate the time required for the activity of each of these isotopes to fall to 1.0% of its initial value. Radioiodine and Radiostrontium tend to concentrate in the thyroid and the bones, respectively of mammals which ingest them. Which isotope is likely to produce the move serious long term effects? ($T_{1/2}$ of ^{131}I = 8 days; $T_{1/2}$ of Sr = 12960 minutes)

5. The half-life of ^{212}Pb is 10.6 hour, that of its daughter ^{212}Bi is 60.5 minute. How long will it take for a maximum daughter activity to grow in freshly seperated ^{212}Pb?

6. While studying the decomposition of gaseous N_2O_5 it is observed that a plot of logarithms of its partial pressure versus time is linear. What kinetic parameters can be obtained from this observation?

7. The progress of the reaction $A \rightleftharpoons nB$, with time is presented in the figure. Determine.
   i) the value of n
   ii) the equilibrium constant, K, and
   iii) the initial rate of conversion of A.



8. One of the hazards of nuclear explosion is the generation of Sr^{90} and its subsequent incorporation in bones. This nuclide has a half life of 28.1 years. Suppose one microgram was absorbed by a new born baby, how much Sr^{90} will remain in his bones after 20 yrs.

9. A painting claimed to be an original of Raphael (1483–1520) is analysed. A small piece of canvas gives C–14 content to be 0.94 of that in the living plants. Is the painting a forgery ($t_{1/2}$ of C^{14} = 5500 years). Assume the present year to be 1999.

10. For the reaction $2NO + Cl_2 \longrightarrow 2NOCl$, it was found that on doubling concentration of both reactants the rate increases eight fold. But on doubling the concentration of chlorine alone, rate only doubles. What is the overall order.

LEVEL – II

1. The rate constant for the first order decomposition of a certain reaction is described by the equation:

$\log K(s^{-1}) = 14.34 - \frac{1.25 \times 10^4}{T}$

i) What is the energy of activation for this reaction

ii) At what temperature will its half-life period be 256 minutes?

2. The rate constant for the second order reaction was shown by $\log k = 12 - \frac{3163}{T}$. Concentration and time were in mol L^{-1} and min. respectively. Calculate the half-life at 43.3°C. The initial concentration of each reactants was 0.001 mol L^{-1}.

3. The gas phase decomposition of dimethyl ether follows first order kinetics

$CH_3–O–CH_3(g) \longrightarrow CH_4(g) + H_2(g) + CO(g)$

The reaction is carried out in a constant volume container at 500°C and has a half life of 14.5 minutes. Initially only dimethyl ether is present at a pressure of 0.4 atm. What is the total pressure of the system after 12 mins. Assume ideal gas behaviour.

4. A sample of pitch blend is found to contain 50% Uranium and 2.425% Lead. Of this Lead only 93% was Pb^{206} isotope. If the disintegration constant is 1.52×10^{-10} yr^{-1}, how old could be the pitch blend deposit.

5. Disintegration of radium takes place at an average rate of 2.24×10^{13} α-particles per minute. Each α-particle takes up 2 electrons from the air and becomes a neutral helium atom. After 420 days, the He gas collected was 0.5×10^{-3} L measured at 300 K and 750 mm of mercury pressure. From the above data calculate Avagadro's number.

6. $A(g) \longrightarrow B(g)$ occurs at a constant temperature. This reaction is taking place simultaneously in two separate vessels. In one vessel where the initial concentration of A(g) was 5 mole/litre the initial rate was 0.035 mole lit^{-1} min^{-1}. But in another vessel where the initial concentration of A(g) was 6 mol/lit after a certain point of time the instantaneous rate was 0.021 moles lit^{-1} min^{-1} when the concentration was 3.2 moles/lit at that point. Calculate the order of the reaction.

7. For a reaction, the energy of activation is zero. What is the value of rate constant at 300 K if $k = 1.6 \times 10^6$ s^{-1} at 280 K? ($R = 8.31$ J K^{-1} mol^{-1}).

8. The first order reaction

Sucrose $\longrightarrow$ Glucose + Fructose

takes place at 308 K in 0.5 N HCl. At zero time, the initial total rotation of the mixture is 32.4°. After 10 minutes, the total rotation is 28.8°. If the rotation of sucrose per mol is 85°, that of Glucose is 74° and fructose is –86.4°. Calculate the half life of the reaction.

9. The time required for 10% completion of first order reaction at 298 K is equal to that required for its 25% completion at 308K. If the preexponential factor for the reaction is 3.56×10^{9} s^{-1}, calculate its rate constant at 318 K and also the energy of activation.

10. A drug becomes ineffective after 30% decomposition. The original concentration of a sample was 5 mg/ml which becomes 4.2 mg/ml during 20 months. Assuming the decomposition of first order and calculate the expiry time of the drug in months. What is the half life period of this product?

11.2 Objective

LEVEL – I

1. A first order reaction is 87.5% complete in an hour. The rate constant of the reaction is
(A) 0.0346 min^{-1} (B) 0.0693 h^{-1}
(C) 0.0693 min^{-1} (D) 0.0346 h^{-1}

2. The half -life a first order reaction is 24 hours. If we start with 10M initial concentration of the reactant then conc. after 96 hours will be
(A) 6.25 M (B) 1.25 M
(C) 0.125 M (D) 0.625M

3. During the particular reaction 10% of the reactant decompose in one hour 20% in two hours 30% in three hours and so on. The unit of the rate constant is
(A) $Hour^{-1}$ (B) L mol^{-1} $hour^{-1}$
(C) mol L^{-1} $hour^{-1}$ (D) mol $hour^{-1}$

4. The temperature coefficient of a reaction is 2, by what factor the rate of reaction increases when temperature is increased from 30°C to 80°C.
(A) 16 (B) 32
(C) 64 (D) 128

5. The rate constant, the activation and Arrhenius parameter of a chemical reaction at 25°C are 3 × 10–4 s–1, 104.4 kJ mol–1 and 6 × 1014 s–1 respectively. The value of the rate constant at $T \to \infty$ is
(A) 2 × 1018s^{-1} (B) 6×10^{14}
(C) α (D) $3.6 \times 10^{30} s^{-1}$

6. At 250°C, the half life for the decomposition of N_2O_5 is 5.7 hour and is independent of initial pressure of N_2O_5. The specific rate constant is
(A) $\frac{0.693}{5.7}$ (B) 0.693×5.7
(C) $\frac{5.7}{0.693}$ (D) None

7. For a given reaction of first order, it takes 20 min, for the concentration to drop from 1 M L^{-1} to 0.6 ML^{-1}. The time required for the concentration to drop from 0.6 ML^{-1} to 0.36 ML^{-1} will be
(A) $>$ 20 min (B) $<$ 20 min
(C) = 20 min (D) ∞

8. In a first order reaction the $\frac{a}{a-x}$ was found to be 8 after 10 min. The rate constant is
(A) $\frac{(2.303 \times 3 \log 2)}{10}$ (B) $\frac{(2.303 \times 2 \log 3)}{10}$
(C) $10 \times 2.303 \times 2 \log 3$ (D) $10 \times 2.303 \times 3 \log 2$

9. For the reaction A + B $\longrightarrow$ Products, it si found that the order of A is 2 and of B is 3 in the rate expression. When concentration of both is doubled the rate will increase by
(A) 10 (B) 6
(C) 32 (D) 16

10. The rate law of the reaction A + 2B $\longrightarrow$ Product is given by d[Product]/dt = K$[A]^2$·[B]. If A is taken in large excess, the order of the reaction will be
(A) 0 (B) 1
(C) 2 (D) 3

11. If a reaction with $t_{1/2}$ = 69.3 sec, has a rate constant 10^{-2} sec^{-1}, the order is
(A) 0 (B) 1
(C) 2 (D) 3

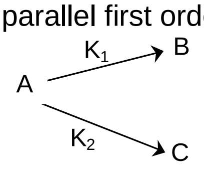
12. The specific rate constant for a first order reaction is 60×10^{-4} sec^{-1}. If the initial concentration of the reactant is 0.01 mol L^{-1}, the rate is
(A) 60×10^{-6} M sec^{-1} (B) 36×10^{-4} M sec^{-1}
(C) 60×10^{-2} M sec^{-1} (D) 36×10^{-1} M sec^{-1}

13. The rate constant for a zero order reaction is 2×10^{-2} mol L^{-1} sec^{-1}. If the concentration of the reactant after 25 sec is 0.5 M, the initial concentration must have been
(A) 0.5 M (B) 1.25 M
(C) 12.5 M (D) 1.0 M

14. A first order reaction is carried out with an initial concentration of 10 ML^{-1} and 80% of the reactant changes into the product. Now if the same reaction is carried out with an initial concentration of 5 ML^{-1}, the percentage of reactant changing to the product is
(A) 40 (B) 80
(C) 160 (D) can't be calculated

15. What fraction of a reactant showing first order remains after 40 min, if $t_{1/2}$ is 20 min?
(A) 1/4 (B) 1/2
(C) 1/8 (D) 1/6

16. A substance undergoes a first order decomposition. The decomposition follows two parallel first order reaction as



The percentage distribution of B and C are
(A) 80% B and 20% C (B) 75% B and 25% C
(C) 90% B and 10% C (D) 60% B and 40% C

17. A tangent drawn on the curve obtained by plotting concentration of product (mole L^{-1}) of a first order reaction vs. time (min) at the point corresponding to time 20 minute makes an angle to 30° with concentration axis. Hence the rate of formations of product after 20 minutes will be
(A) 0.580 mole L^{-1} min^{-1} (B) 1.723 mole L^{-1} min^{-1}
(C) 0.290 mole L^{-1} min^{-1} (D) 0.866 mole L^{-1} min^{-1}

18. For reaction 3A $\longrightarrow$ products, it is found that the rate of reaction increases 4- fold when concentration of A is increased 16 times keeping the temperature constant. The order of reaction is?
(A) 2 (B) 1
(C) 1 (D) 0.5

19. The thermal decomposition of acetaldehyde : $CH_3CHO \longrightarrow CH_4 + CO$, has rate constant of 1.8×10^{-3} $mole^{-1/2}L^{1/2}$ min^{-1} at a given temperature. How would $-\frac{d[CH_3CHO]}{dt}$ will change if concentration of acetaldehyde is doubled keeping the temperature constant?
(A) will increase by 2.828 times (B) will increase by 11.313 times
(C) will not change (D) will increase by 4 times

20. The reaction ; $2O_3 \longrightarrow 3O_2$, is assigned the following mechanism.
I) $O_3 \rightleftharpoons O_2 + O$
II) $O_3 + O \xrightarrow{\textbf{slow}} 2O_2$
The rate law of if the reaction will, therefore be
(A) $r \propto [O_3]^2[O_2]$ (B) $r \propto [O_3]^2 [O_2]^{-1}$
(C) $r \propto [O_3]$ (D) $r \propto [O_3] [O_2]^{-2}$

21. For an endothermic reaction where ΔH represents the enthalpy of the reaction, the minimum value for the energy of activation will be
(A) Less than ΔH (B) zero
(C) more than ΔH (D) equal to ΔH

22. The rate expression for a reaction is rate = $k[A]^{3/2} [B]^{-1}$, the order of reaction is
(A) 0 (B) 1/2
(C) 3/2 (D) 5/2

23. The reaction $A(g) + 2B(g) \longrightarrow C(g) + D(g)$ is an elementary process. In an experiment, the initial partial pressure of A & B are $P_A = 0.60$ and $P_B = 0.80$ atm. When $P_C = 0.2$ atm the rate of reaction relative to the initial rate is
(A) 1/48 (B) 1/24
(C) 9/16 (D) 1/6

24. If concentration are measured in mole/lit and time in minutes, the unit for the rate constant of a 3rd order reaction are
(A) mol lit^{-1} min^{-1} (B) lit^2 $mol^{-2}min^{-1}$
(C) lit.mol^{-1} min^{-1} (D) min^{-1}

25. A radioactive element has a half life period of 140 days. How much of it will remain after 1120 days

(A) $\frac{1}{32}$ (B) $\frac{1}{250}$

(C) $\frac{1}{512}$ (D) $\frac{1}{128}$

26. For a first order reaction the plot of log $[A]_t$ Vs t is linear with a

(A) positive slope and zero intercept
(B) positive slope and non zero intercept
(C) negative slope and zero intercept
(D) negative slope and non zero intercept

27. For a hypothetical reaction A + B $\longrightarrow$ C+D, the rate = $k[A]^{-1/2}[B]^{3/2}$. On doubling the concentration of A and B the rate will be

(A) 4 times (B) 2 times
(C) 3 times (D) none of these

28. In a reaction the threshold energy is to equal to

(A) average energy of the reactants
(B) activation energy
(C) activation energy + average energy of the reactants
(D) activation energy – average energy of the reactants

29. For a first reaction $t_{0.75}$ is 138.6 seconds. Its specific rate constant (in sec^{-1}) is

(A) 10^{-2} (B) 10^{-4}
(C) 10^{-5} (D) 10^{-6}

30. Half life period for a first–order reaction is 10 minutes. How much time is required to change the concentration of the reactants from 0.08M to 0.01M

(A) 20 min (B) 30 min
(C) 40 min (D) 50 min

LEVEL - II	More than one choice

1. Which of the following statements are correct?

(A) The order of a reaction is the sum of the components of all the concentration terms in the rate equation.
(B) The order of a reaction with respect to one reactant is the ratio of the change of logarithm of the rate of the reaction to the change in the logarithm of the concentration of the particular reactant, keeping the concentrations of all other reactants constant.
(C) Orders of reactions can be whole numbers of fractional numbers.
(D) The order of a reaction can only be determined from the stoichiometric equation for the reaction.

2. Which of the following statements are correct?

(A) The rate of the reaction involving the conversion of ortho-hydrogen to parahydrogen is $-\frac{d[H_2]}{dt} = k[H_2]^{3/2}$
(B) The rate of the reaction involving the thermal decomposition of acetaldehyde is $k[CH_3CHO]^{3/2}$
(C) In the formation of phosgene gas from CO and Cl_2, the rate of the reaction is $k[CO][Cl_2]^{1/2}$
(D) In the decomposition of H_2O_2, the rate of the reaction is $k[H_2O_2]$

3. Which of the following reactions is of the first order?
(A) The decomposition of ammonium nitrate in an aqueous solution
(B) The inversion of cane-sugar in the presence of an acid
(C) The acidic hydrolysis of ethyl acetate
(D) All radioactive decays

4. The calculation of the pre-exponential factor is based on the
(A) idea that, for a reaction to take place, the reactant species must come together
(B) calculation of the molecularity of the reaction
(C) idea that the reactant species must come together, leading to the formation of the transition state which then transforms into the products
(D) calculation of the order of the reaction

5. The basic theory behind Arrhenius's equation is that
(A) the number of effective collisions is proportional to the number of molecules above a certain threshold energy
(B) as the temperature increases, so does the number of molecules with energies exceeding the threshold energy
(C) the rate constant is a function of temperature
(D) the activation energy and pre-exponential factor are always temperature-independent

6. In Arrhenius's equation, k = A exp $\left(-\frac{E_a}{RT}\right)$. A may be termed as the rate constant at
(A) very low temperature (B) very high temperature
(C) zero activation energy
(D) the boiling temperature of the reaction mixture

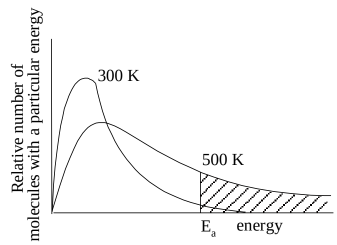
7. Rate constant k varies with temperature by eqn:
log k (min^{-1}) = 5– $\frac{2000\ K}{T}$. We can conclude
(A) pre-exponential factor A is 5 (B) E_a is 2000 kcal
(C) pre-exponential factor A is 10^5 (D) E_a is 9.212 kcal

8. A reaction is catalysed by H^+ ion. In presence of HA, rate constant is 2×10^{-3} min^{-1} and in presence of HB rate constant is 1×10^{-3} min^{-1}, HA and HB both being strong acids, we may conclude:
(A) equilibrium constant is 2
(B) HAis stronger than HB

(C) relative strength of HA and HB is 2
(D) HA is weaker than HB and relative strength is 0.5

9. Which of the following statements are correct?
(A) A plot of log K_p versus 1/T is linear
(B) A plot of log [X] versus time is linear for a first order reaction $X \longrightarrow P$
(C) A plot of log P versus 1/T is linear at constant volume
(D) A plot of P versus 1/V is linear at constant temperature

10. The distribution of molecular kinetic energy at two temperatures is as shown in the following graph.



Which of the following conclusions are correct?
(A) The number of molecules with energy E_a or greater is proportional to the shaded area for each temperature
(B) The number of molecules with energy E_a or less is proportional to the shaded area for each temperature.
(C) The number of molecules with energy E_a is the mean of all temperatures
(D) The graph follows the Maxwell-Boltzmann energy distribution law.

LEVEL - III	Other Engg. Exams

1. For the reaction $2N_2O_5 \rightarrow 4NO_2 + O_2$ rate of reaction and rate constant are 1.02×10^{-4} and 3.4×10^{-5} sec^{-1} respectively. The concentration of N_2O_5 at that time will be
(A) 1.732
(B) 3
(C) 1.02×10^{-4}
(D) 3.4×10^{5}

2. A reaction involving two different reactants
(A) Can never be a second order reaction
(B) Can never be a unimolecular reaction
(C) Can never be a bimolecular reaction
(D) Can never be a first order reaction

3. What is the order of a reaction which has a rate expression rate = $K[A]^{3/2}[B]^{-1}$
(A) 3/2
(B) 1/2
(C) 0
(D) none of these

4. Which of the following expression is correct for first order reaction ? (CO) refers to initial concentration of reactant
(A) $t_{1/2} \propto CO$
(B) $t_{1/2} \propto CO^{-1}$
(C) $t_{1/2} \propto CO^{-2}$
(D) $t_{1/2} \propto CO^{0}$

5. Which of the following statements about zero order reaction is not true
(A) Its unit is sec^{-1}
(B) The graph between log (reactant) versus rate of reaction is a straight line

(C) The rate of reaction increases with the increase in concentration of reactants
(D) Rate of reaction is independent of concentration of reactants

6. For the reaction A + 2B $\rightarrow$ C, rate is given by R = [A] $[B]^2$ then the order of the reaction is
(A) 3 (B) 6
(C) 5 (D) 7

7. Units of rate constant of first and zero order reactions in terms of molarity M unit are respectively
(A) sec^{-1}, M sec^{-1} (B) sec^{-1}, M
(C) M sec^{-1}, sec^{-1} (D) M, sec^{-1}

8. For the reaction system 2NO (g) + O_2 (g) $\rightarrow$ 2NO_2 (g) volume is suddenly reduced to half its value by increasing the pressure on it. If the reaction is of first with respect to O_2 and second order with respect to NO, the rate of reaction will
(A) Diminish to one fourth of its initial value
(B) Diminish to one eighth of its initial value
(C) Increase to eight times of its initial value
(D) Increase to four times of its initial value

9. In the first order reaction, the concentration of the reactant is reduced to 25% in one hour. The half life period of the reaction is
(A) 2hr (B) 4hr
(C) 1/2 hr (D) 1/4 hr

10. For a reaction, X (g) $\rightarrow$ Y(g) + Z(g) the half life period is 10 min. In what period of time would the concentration of X be reduced to 10% of original concentration
(A) 20 min (B) 33 min
(C) 15 min (D) 25 min

11. A radioactive isotope having a half-life of 3 days was received after 12 days. It was found that there were 3 gm of the isotope in the container. The initial weight of the isotope when packed was
(A) 12 gm (B) 24 gm
(C) 36 gm (D) 48 gm

12. Half-life period of a first order reaction is
(A) Inversely proportional to the concentration
(B) Independent of the concentration
(C) Directly proportional to the initial concentration
(D) Directly proportional to the final concentration

13. A radioactive element resembling iodine in properties is
(A) Astatine (B) Lead
(C) Radium (D) Thorium

14. The C^{14} to C^{12} ratio in a wooden article is 13% that of the fresh wood. Calculate the age of the wooden article. Given that the half-life of C^{14} is 5770 yers.
(A) 16989 years (B) 16858 years
(C) 15675 years (D) 17700 years

15. Hydrogen bomb is based on the principle of

(A) Nuclear fission (B) Natural radioactivity
(C) Nuclear fusion (D) Artificial radioactivity

LEVEL - IV

A Matrix-Match Type

1. Matching (For I[st]-order reaction)

Column – I		Column – II	
(A)	$t_{63/64}$	(p)	$6t_{1/2}$
(B)	$t_{15/16}$	(q)	$2t_{3/4}$
(C)	$t_{31/32}$	(r)	$\frac{5}{3}t_{7/8}$
(D)	$t_{255/256}$	(s)	$2t_{15/16}$

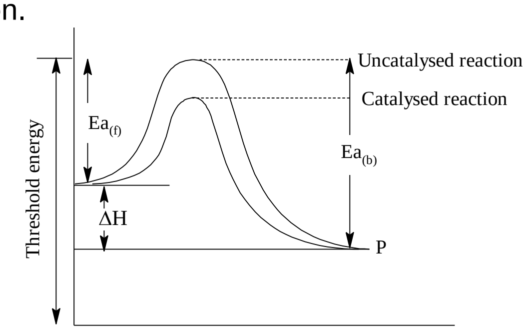
2. Match the following :

Column I		Column II	
(A)	${}_7N^{14} + {}_2He^4 \longrightarrow {}_8O^{17} + ...$	(p)	${}_0n^1$
(B)	$2\,{}_1H^3 \longrightarrow {}_2He^4 + 2......$	(q)	${}_1H^1$
(C)	${}_{94}Pu^{239} + {}_2He^4 \longrightarrow {}_{96}Cm^{242} + ...$	(r)	${}_2He^4$
(D)	${}_{29}Cu^{53} \longrightarrow {}_{28}Ni^{53} + ...$	(s)	${}_1e^0$

B Linked Comprehension Type

C_{1-3} : Paragraph for Questions Nos. 1 to 3

Following is the graph which represents the energy profile diagram for the catalysed and uncatalysed first order reaction. The graph of lower energy barrier is the energy profile diagram for the catalysed reaction and of higher energy barrier is the energy profile diagram of uncatalysed reaction.



The reaction which occurs by absorption of heat is known endothermic reaction and by release of heat or energy is known as exothermic reaction. ΔH is known as change in enthalpy of reaction. For exothermic process, ΔH is negative and for endothermic process, ΔH is positive.

From Arrhenius equation, we know that

$K = Ae^{-\frac{Ea}{RT}}$, where K is rate constant

A is frequency factor or Arrhenius constant or maximum rate constant.

The catalysed reaction occurs at 300K and uncatalysed reaction occurs at 400K in such a way that the rate of catalysed reaction and rate of uncatalysed reaction becomes equal as well as the catalyst lower down the energy barrier by 20J.

1. What will be the value of activation energy of the uncatalysed reaction?
(A) 40J (B) 60J
(C) 30J (D) 80J

2. From the above information given, what will be the threshold energy of the catalysed reaction, if the normal energy of the reactant is 50J ?
(A) 100J (B) 120J
(C) 80J (D) 110J

3. From the above information given, if the normal energy of reactant is 40J, and product is 20J then what will be the activation energy of the uncatalysed backward reaction?
(A) 85J (B) 90J
(C) 110J (D) 100J

$C_{4–6}$: Paragraph for Questions Nos. 4 to 6

Ozone is prepared in laboratory by passing silent electric discharge through pure and dry oxygen in an apparatus known as ozoniser. This conversion from oxygen to ozone is a reversible and endothermic reaction. When oxygen is subjected to an ordinary electric discharge, most of the O_3 produced will get decomposed. When any insulating material such as glass, is inserted in the space between the two electrodes and high current density is applied, silent electric discharge passes on between the two electrodes. By this process no spark is produced and much less heat is generated, and as a result the decomposition of the produced ozone is much retarded.

The decomposition of ozone is believed to occur by the following mechanism:

$O_3 \underset{K_b}{\overset{K_f}{\rightleftharpoons}} O_2 + O$ (fast step)

$O_3 + O \xrightarrow{K_1} 2O_2$ (slow)

4. Order of the reaction is
(A) 1 (B) 2
(C) 3 (D) 0

5. Molecularity of reaction is defined by
(A) slow step (B) reversible step
(C) from overall reaction (D) fast step

6. When the concentration of O_3 is increased, for the same concentration of oxygen, its rate
(A) increases (B) decreases
(C) remains the same (D) cannot be answered

Rate of forward reaction decreases and that of backward reaction increases with passage of time. The rate of reaction increases considerable with an increase in temperature.

$C_{7–9}$: Paragraph for Questions Nos. 7 to 9

7. The rate of a chemical reaction generally increases rapidly even for small temperature increase because of rapid increase in the
(A) collision frequency (B) fraction of molecules
(C) activaton energy (D) average kinetic energy of molecules

8. The temperature coefficient of a reaction is:
(A) ratio of rate constants at two temperatures differing by 1°C
(B) ratio of rate constants at temperature 35°C and 25°C
(C) ratio of rate constants at temperature 30°C and 25°C
(D) specific reaction rate at 25°

9. The specific rate constant of a first order reaction depends on
(A) concentration of the reactants (B) concentration of products
(C) times (D) temperature

16. Hints to Subjective Assignments

LEVEL – I

1. Apply, $K = \frac{0.693}{t_{1/2}}$ and $\log\frac{K_2}{K_2} = \frac{Ea}{2.303R}\left[\frac{T_2 - T_1}{T_1 \times T_2}\right]$

5. Apply $t_{max} = \frac{2.303}{\lambda_d - \lambda_p}\log\frac{\lambda_d}{\lambda_p}$

10. Rate = $Rate = K \times [A]^x [B]^y$
Calculate the value of x and y using given data then calculate order.

LEVEL – II

1. For second order reaction
$K = \frac{1}{t}\frac{x}{(a - x)}$

8. Apply $K = \frac{2.303}{t}\log\frac{r_0 - r_\infty}{r_t - r_\infty}$

13. Answers

13.1 Subjective

LEVEL – I

1.	128.33 hours	2.	$k_2 = 0.35$ mol^{-1} dm^3 s^{-1}
3.	20.33 min	4.	59.8 days
5.	227 min	7.	1.2, 0.1 M hr^{-1}
8.	N = 6.1×10^{-7}g	9.	491.16 year
10.	3		

LEVEL – II

1.	669.16K	2.	10 minutes
3.	0.75 atm	4.	3.3×10^6 years
5.	6.775×10^{23}	6.	1.128
7.	1.6×10^6 s^{-1}	8.	61.16 min
9.	9.036×10^{-4} s^{-1}	10.	80 months

13.1 Objective

LEVEL – I

1.	**A**	2.	**D**
3.	**C**	4.	**B**
5.	**B**	6.	**A**
7.	**C**	8.	**A**
9.	**C**	10.	**B**
11.	**B**	12.	**A**
13.	**A**	14.	**B**
15.	**D**	16.	**B**
17.	**B**	18.	**D**
19.	**A**	20.	**D**
21.	**C**	22.	**B**
23.	**D**	24.	**B**
25.	**A**	26.	**A**
27.	**B**	28.	**B**
29.	**C**	30.	**A**

LEVEL – II

1.	(A, B, C)	2.	(A, B, C, D)
3.	(A, B, C, D)	4.	(A, C)
5.	(A, B, C)	6.	(B, C)
7.	(C, D)	8.	(B, C)
9.	(A, B, D)	10.	(A, D)

LEVEL – III

1.	(B)	2.	(B)
3.	(B)	4.	(B)
5.	(D)	6.	(B)
7.	(A)	8.	(A)
9.	(C)	10.	(B)
11.	(D)	12.	(B)
13.	(A)	14.	(A)
15.	(C)		

LEVEL – IV

A		B	
Matrix-Match Type		**Linked Comprehension Type**	
1.	(A) – (p)	1.	(D)
	(B) – (q)	2.	(D)
	(C) – (r)	3.	(D)
	(D) – (s)	4.	(B)
2.	(A) – (q)	5.	(A)
	(B) – (p)	6.	(A)
	(C) – (p)	7.	(B)
	(D) – (s)	8.	(D)
		9.	(B)

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